MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
1800 SECURITY BASE CAMP: MOSQUE
DESIGN OF BUILDING STRUCTURE

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
Typ.	:	Typical
HCS	:	Hollow Core Slab
Fc'	:	Concrete Compressive Strength
Fy	:	Yield Strength of Reinforcing Steel
ϕ	:	Strength Reduction Factor
SBC	:	Saudi Building Code
SASO	:	Saudi Standards, Metrology and Quality Organization
ACI	:	American Concrete Institute
ASCE	:	American Society of Civil Engineers
PCI	:	Precast/Prestressed Concrete Institute

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1. INTENT

This Document is intended to produce the technical needs in Engineering including particular reference to design justification & Engineering validation works. The work described below is related to Design of a Mosque Building in Security Base Camp area.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The Scope of this document is limited to Complete Analysis & Structural Design Verification of Mosque Building only.

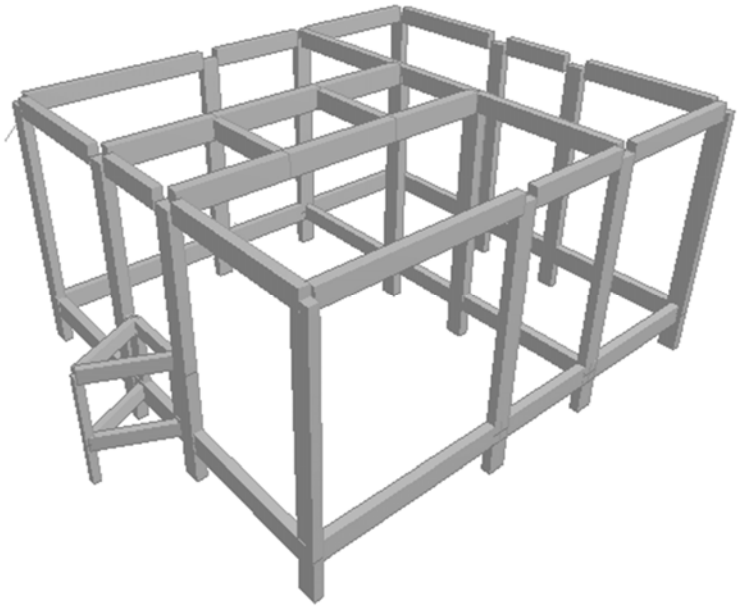
4. DESIGN METHODOLOGY & ASSUMPTIONS

This Building Is a Single Story Frame Structure with Foundation Being Cast in Site. The Building Shall Have Insulated Block Walls and Cast in Site Framing Beams to support the HCS (Hollow Core Slab)The Superstructure for this Building was modeled using STAAD software. The Analysis of Frames (Cast-In-Situ and Steel Columns & Beams) has been performed meeting design requirements & keeping in view the Architectural requirements. The loads were applied as per SBC/ASCE/ACI standards & analysis was done to generate the Column Reactions at the Base that are assumed to be fixed. The Foundation design has been done by importing Column Base Reactions into Foundation Design Software.

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STAAD 3D MODEL

4.1. Codes

[1] SBC-301-2007- Structural Loading & Forces.

[2] SBC-303-2007- Structural Soil & Foundations.

[3] SBC-304-2007- Structural Concrete Structures.

[4] SASO SSA 2 - Steel Bars for the Reinforcement of Concrete.

[5] SASO SSA 224 - Steel Fabric for Reinforcement of Concrete

[6] ASCE-7-2005 - Minimum Design Loads for Buildings and other Structures.

[7] ACI 318-08 - Building Code Requirements.

[8] PCI –MNL-120-04 - Design Handbook, 6th edition.

[9] SBC-306-2007- Structural – Steel Structures.

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4.2. Geotechnical investigations

Foundations have been designed based on recommendations of the Soil Investigation Report, Doc # MD-216-3000-EG-CV-RPT-1008.

4.3. Material Properties

Compressive Strength of Concrete at 28th days

fc' = 35 MPa (5.00 Ksi) for beams, columns, suspended slabs.

fc' = 35 MPa (5.00 Ksi) for foundations, grade beams, columns.

Yield Strength of Steel

fy = 414 MPa (60 Ksi) for all reinforcing steel.

5. DESIGN LOADS

5.1. Dead Load(DL)

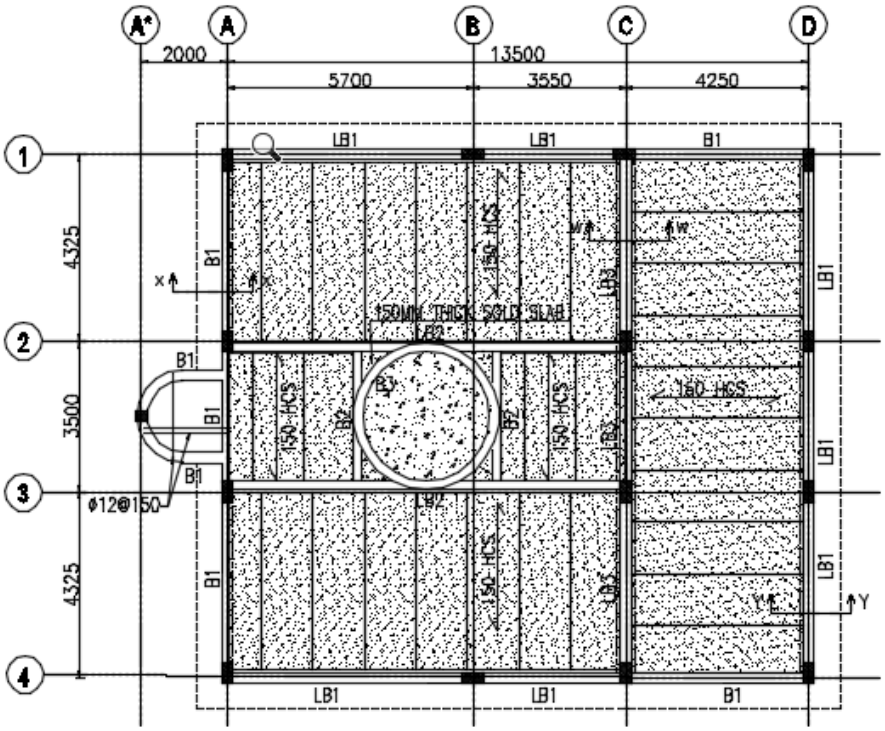
5.1.1. Roof Dead Loads

• Provision for Electrical Lighting	= 0.05 kN/m²
Ceiling	
Suspended steel channel system	= 0.10 kN/m²
Acoustical Fiber Tile (16mm)	= 0.05 kN/m²
Sub-total	= **0.2 kN/m²**

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- Hollow Core Slab

  (HCS-150)(To be confirmed by Vendor) b/w Grid Line's (1 & 4), (A & D) = **2.52 KN/m2**

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Prestressed Hollowcore Floor										
Overall Structural Depth (mm)	Self Wt- KN/M2	No. of Cores in Unit	Spans indicated below allow for Characteristic Service Load (Live Load) Plus Self Weight Plus 1.5Kn/m^2 for Finishes Characteristic Service Loads Kn/m^2							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.00
			Clear Span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

- Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = **3.1 kN/ m²**

Therefore

Total Dead Load =3.1+0.2+2.52= 5.82 **kN/m²**

TOTAL DL as UDL ON BEAMS

- UDL on LB's Along Grid Line-1 & 4= 4.25 x 5.82/2 = **13.00 Kn/m**
- UDL on LB's Along Grid Line-2 & 3=13.00+(5.82x 3.00)/2 =**21.8 Kn/m**
- UDL on LB's Along Grid Line's-C, D =4.25 x 5.82/2 =**12.36 Kn/m**

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+ (0.05+0.2)/2x20) *0.6*0.6/2 = **1 knm/m**

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5.1.2. Ground Floor Walls

- 250 thk. Outer CMU (Partially Grouted Cell@800 c-c)

16 x 0.25 x 5.8m clear height = **23.2 kN/m**

5.1.3. Dome Load on Beams

Thickness of Dome = 150mm

Diameter of the Dome = 2.4m

Height of the Dome = 1m

Weight of the Dome = 2.7 x 24 = 65 Kn

Udl of dome on the Supporting Beams = 65/10 = 6.5 Kn/m

5.1.4. Roof Live Loads (Lr)

Live loads are with reference to Clause-4.2.2, MD-216-1000-EG-ST-CRT-1001

Ordinary Flat (Inaccessible) = **0.5 kN/m²**

TOTAL Lr as UDL ON BEAMS

- UDL on LB's Along Grid Line-1 & 4= 4.25 x 0.50/2 = **1.125 Kn/m**
- UDL on LB's Along Grid Line-2 & 3=1.125+(0.50x 3.00)/2 =**1.875 Kn/m**
- UDL on LB's Along Grid Line's-C, D =4.25 x 0.50/2 =**1.0625 Kn/m**

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5*0.6*0.6/2 = **0.09 kN/m /m**

5.2. Wind Load (WL)

Wind loads were automatically generated in STAAD as per ASCE-07. The parameters used for the wind load generation are as follows (Refer STRUCTURAL DESIGN CRITERIA NON-PROCESS BUILDINGS Doc No MD-216-0000-EG-ST-CRT-0001)

All structures and foundations (except for buried structures) designed and shall be constructed to resist the wind effects determined in accordance with Section-7.2, SBC-301-Analytical Procedure.

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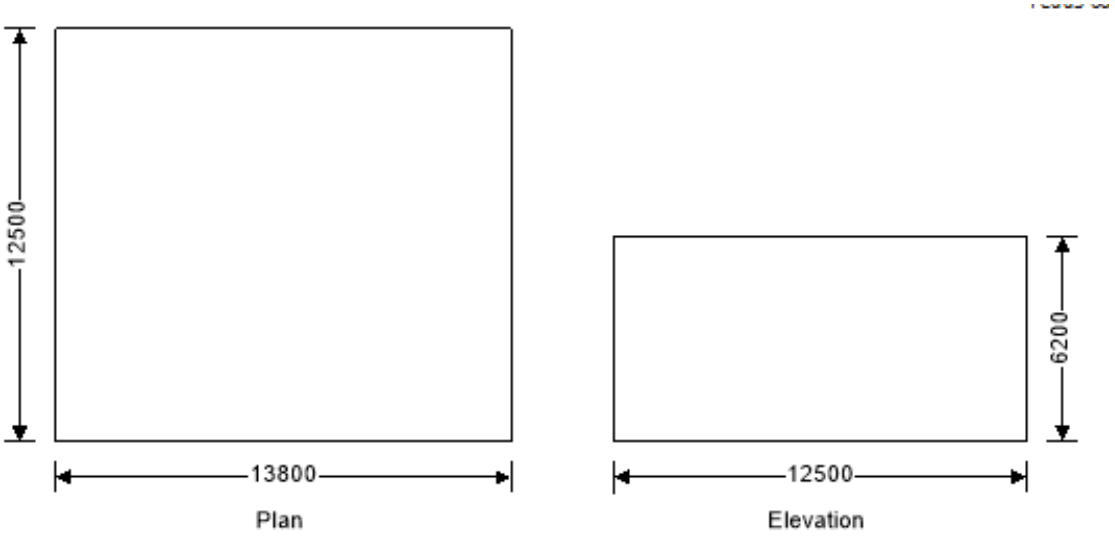
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5.3. Wind Load (Analytical Verification)

5.3.1. WIND LOADING (SBC-301)-MOSQUE AREA

Using the directional design method



Building data

Type of roof;	Flat
Length of building;	b = 13800 mm
Width of building;	d = 12500 mm
Height to eaves;	H = 6200 mm
Mean height;	h = 6200 mm

General wind load requirements

Basic wind speed;	V = **49** m/s (or) 177 kmph (Refer Clause-4.2.6, Doc- MD-216-1000-EG-ST- CRT-1001)
Risk category;	II(As per SBC-301, Table 1.6-1)
Velocity pressure exponent coeff ;	K_d = **0.85**(As per SBC-301, Table 6.4-1)
Exposure category;	D(As per SBC-301, Table 6.4.2.3)

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Enclosure classification; Partially enclosed buildings(As per SBC-301, Section-6.2)

Internal pressure coef +ve; GC_{pi_p} = **0.55**(As per SBC-301, Figure-7.2-1)

Internal pressure coef –ve; GC_{pi_n} = **-0.55**(As per SBC-301, Figure-7.2-1)

Gust effect factor; G_f = **0.85**(As per SBC-301, Section-7.2.7.1)

Importance Factor; 1(As per SBC-301, Section-9.5)

Topography

Topography factor not significant; K_{zt} = 1.0(As per SBC-301, Section-6.4.3.2)

Velocity pressure equation; $q = 0.0473x10^{-3}Kn/m^2 \times K_z \times K_{zt} \times K_d \times V^2$ xI

V in Km/hr(As per SBC-301, Section-7.2.8)

Velocity pressures table

z (mm)	Kz (Table 27.3-1)	qz (kN/m2)
4572	**1.03**	**1.29**
4572	**1.03**	**1.29**
6200	**1.08**	**1.35**
6200	**1.08**	**1.35**

Pressures and forces

Net pressure; $p = q \times G_f \times C_{pe} - q_i \times GC_{pi}$;

Net force; $F_w = p \times A_{ref}$;

Walls load case 1 - Wind 0, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A_1	4.57	0.80	1.29	1.62	63.09	102.29
A_2	4.57	0.80	1.29	1.62	0.00	0.00
A_3	6.20	0.80	1.35	1.67	22.46	37.42
B	6.20	-0.50	1.35	0.17	85.56	14.49
C	6.20	-0.70	1.35	-0.06	77.50	-4.72
D	6.20	-0.70	1.35	-0.06	77.50	-4.72

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Walls load case 2 - Wind 0, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.29	1.62	63.09	102.29
A_2	4.57	0.80	1.29	1.62	0.00	0.00
A_3	6.20	0.80	1.35	1.67	22.46	37.42
B	6.20	-0.50	1.35	0.17	85.56	14.49
C	6.20	-0.70	1.35	-0.06	77.50	-4.72
D	6.20	-0.70	1.35	-0.06	77.50	-4.72

Walls load case 3 - Wind 90, GC_{pi} 0.55, - c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m^2)	Net pressure p (kN/m^2)	Area A_{ref} (m^2)	Net force F_w (kN)
A_1	4.57	0.80	1.29	0.13	57.15	7.50
A_2	4.57	0.80	1.29	0.13	0.00	0.00
A_3	6.20	0.80	1.35	0.18	20.35	3.58
B	6.20	-0.48	1.35	-1.30	77.50	-100.50
C	6.20	-0.70	1.35	-1.55	85.56	-132.70
D	6.20	-0.70	1.35	-1.55	85.56	-132.70

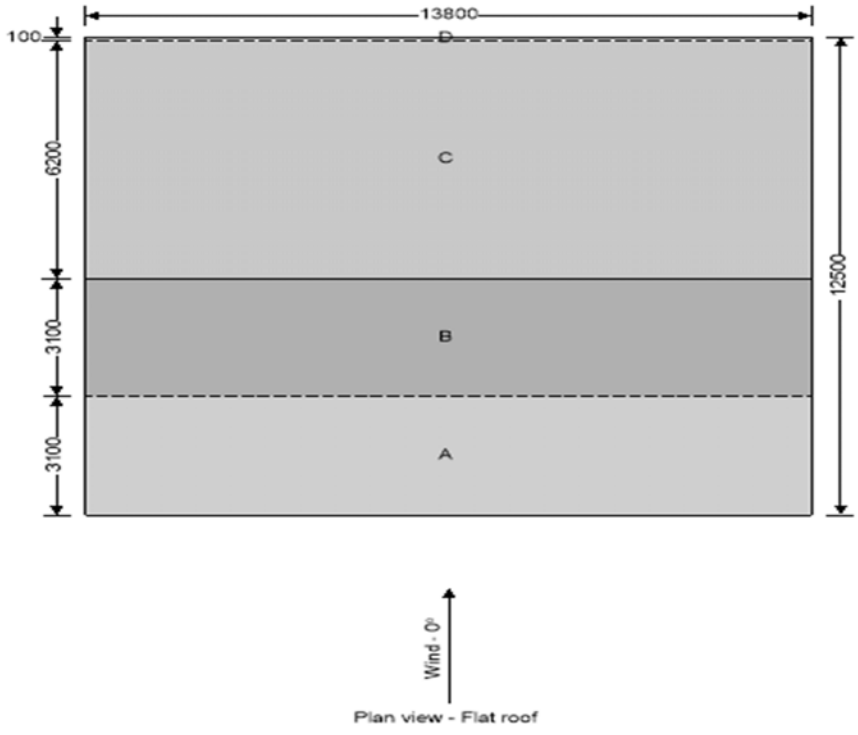
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Walls load case 4 - Wind 90, GC_{pi} -0.55, + c_{pe}

Zone	Ref. height (m)	Ext pressure coefficient c_{pe}	Peak velocity pressure q_p (kN/m²)	Net pressure p (kN/m²)	Area A_{ref} (m²)	Net force F_w (kN)
A_1	4.57	0.80	1.29	1.62	57.15	92.65
A_2	4.57	0.80	1.29	1.62	0.00	0.00
A_3	6.20	0.80	1.35	1.67	20.35	33.90
B	6.20	-0.48	1.35	0.19	77.50	14.98
C	6.20	-0.70	1.35	-0.06	85.56	-5.22
D	6.20	-0.70	1.35	-0.06	85.56	-5.22

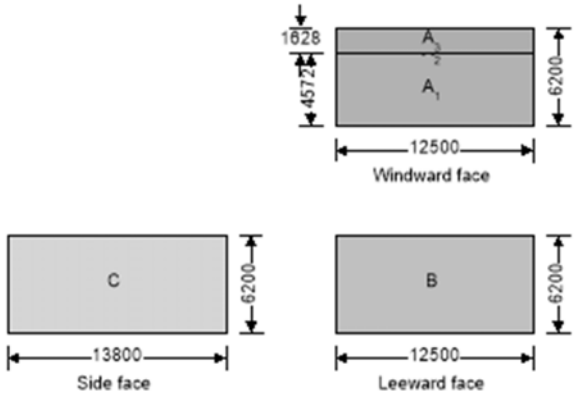


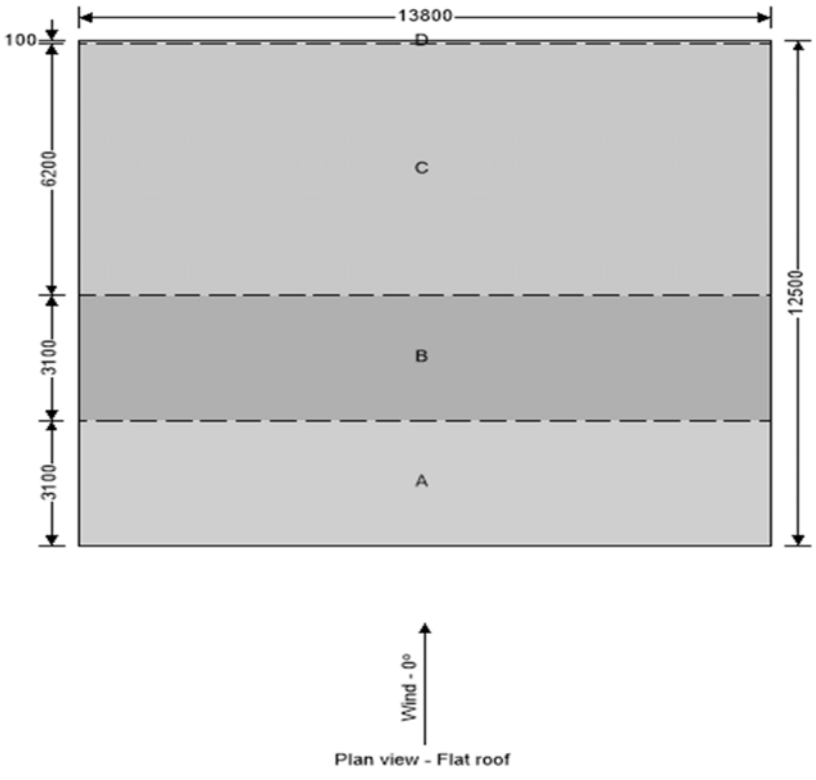
Plan view - Flat roof

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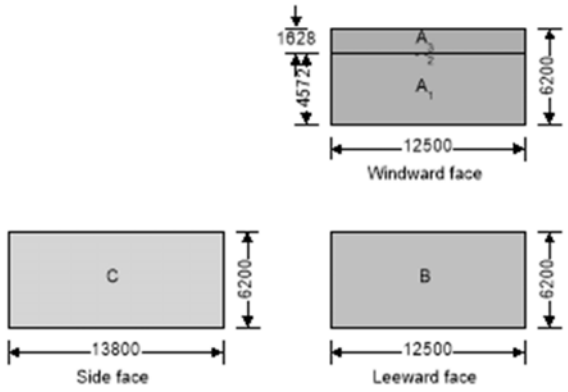




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So from the above analysis, the Wind Loads generated in STAAD are in accordance with SBC-301-2007-Structural Loading & Forces.

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5.4. Seismic Load(E)

Earthquake loads were automatically generated in STAAD as per IBC 2003. The basic parameters affecting the generation of seismic loads is shown below:

Importance Factor, I = 1.0(As per SBC-301, Section-9.5)

Design Spectral Response Acceleration (Refer Clause-4.2.7b, MD-216-1000-EG-ST- CRT-1001)

Ss = 0.1g (Spectral Response Acceleration 0.2 second)

S1 = 0.02g (Spectral Response Acceleration 1 second)

S_{D_1} = 2/3*Fv*S1 = 2/3 * 2.4 * 0.02 = 0.032 (As per SBC-301, Section-9.4.4)

S_{DS} = 2/3*Fa*Ss = 2/3 * 1.6 * 0.1 = 0.106 (As per SBC-301, Section-9.4.4)

Where S_{D_1} Damped Design Spectral Response Acceleration@1 Sec

S_{DS} Damped Design Spectral Response Acceleration@0.2 Sec

Site Class = D

Site Coefficient, Fa = 1.60 (As per SBC-301, Table 9.4.3a)

Site Coefficient, Fv = 2.40 (As per SBC-301, Table 9.4.3b)

Building period Coefficient, Ct=0.04 (As per SBC-301, Table 10.9.3.2)

For Ordinary reinforced concrete moment frame,

Response Modification Factor, Rx = 2.5 (As per SBC-301, Table 10.9.3.2)

Response Modification Factor, Rz = 2.5 (As per SBC-301, Table 10.2)

System Over Strength Factor, Omega = 3 (As per SBC-301, Table 10.2)

Deflection Amplification factor, Cd = 2.5 (As per SBC-301, Table 10.2)

5.5. Thermal/Temperature Loads(Self-restraining ,Thermal Expansion & Contraction Loads)

An ambient temperature ranges from a maximum of 50° C to a low of 0° C. In direct exposure to sunlight, the temperature shall be assumed to rise a minimum of 15° C above ambient. As a minimum, the temperature gradient for thermal loads shall be +25° C and -15° C with the most severe condition governing the design.

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6. DESIGN COMBINATIONS

Service Loads (Allowable Stress Design) (SBC-301, Section 2.4.1)

D

D + T

D +Lr

D +0.75Lr

D + (W or 0.7E)

D + 0.75W + 0.75Lr

0.6D + W

0.6D + 0.7E

Factored loads (Strength Design) (SBC-301, Section 2.3.2)

1.4D

1.4D+1.4T

1.2(D + T) + 0.5Lr

1.2D + 1.6Lr + 0.8W

1.2D + 1.6W + 0.5Lr

1.2D + 1.0E

0.9D + 1.6W

0.9D + 1.0E

Where:

D = Dead Load

Lr = Live Load

W = Wind Load

E = Seismic Load

T = Temperature Load

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7. CRACKED SECTIONS

In lieu of a more accurate analysis of the effects of cracking of concrete, provisions of SBC-304-Section 10.11 was used.

For Beams:	I = 0.35Ig
For Columns:	I = 0.70Ig
For Walls:	
Un-cracked	I = 0.70Ig
Cracked	I = 0.35Ig
For Slabs	I = 0.25Ig

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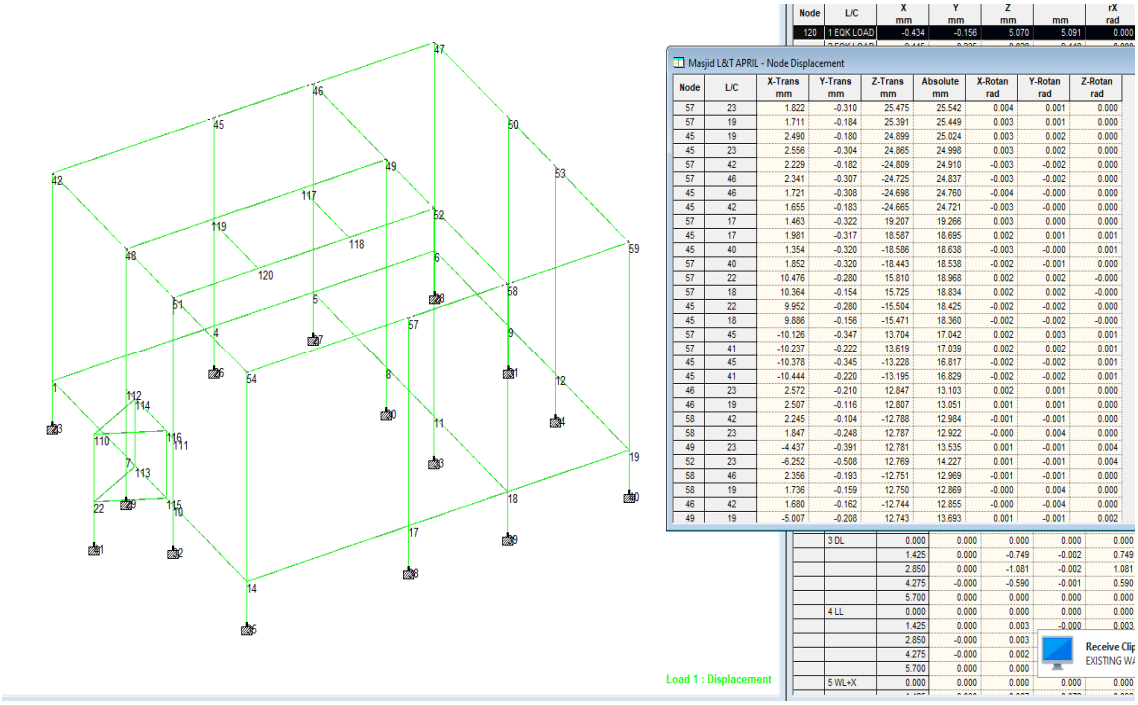
8. DEFLECTIONS

8.1. Maximum Allowable Deflection

L/240: for member load with live load plus dead load

L/600: for members supporting masonry under total loads

Where, L = span of member between support



Masjid L&T APRIL - Node Displacement

Node	L/C	X-Trans mm	Y-Trans mm	Z-Trans mm	Absolute mm	X-Rotan rad	Y-Rotan rad	Z-Rotan rad
57	23	1.822	-0.310	25.475	25.542	0.004	0.001	0.000
57	19	1.711	-0.184	25.391	25.449	0.003	0.001	0.000
45	19	2.490	-0.180	24.899	25.024	0.003	0.002	0.000
45	23	2.556	-0.304	24.865	24.998	0.003	0.002	0.000
57	42	2.229	-0.182	-24.809	24.910	-0.003	-0.002	0.000
57	46	2.341	-0.307	-24.725	24.837	-0.003	-0.002	0.000
45	46	1.721	-0.308	-24.698	24.760	-0.004	-0.000	0.000
45	42	1.655	-0.183	-24.665	24.721	-0.003	-0.000	0.000
57	17	1.463	-0.322	19.207	19.266	0.003	0.000	0.000
45	17	1.981	-0.317	18.587	18.695	0.002	0.001	0.001
45	40	1.354	-0.320	-18.586	18.638	-0.003	-0.000	0.001
57	40	1.852	-0.320	-18.443	18.538	-0.002	-0.001	0.000
57	22	10.476	-0.280	15.810	18.968	0.002	0.002	-0.000
57	18	10.364	-0.154	15.725	18.834	0.002	0.002	-0.000
45	22	9.952	-0.280	-15.504	18.425	-0.002	-0.002	0.000
45	18	9.886	-0.156	-15.471	18.360	-0.002	-0.002	-0.000
57	45	-10.126	-0.347	13.704	17.042	0.002	0.003	0.001
57	41	-10.237	-0.222	13.619	17.039	0.002	0.002	0.001
45	45	-10.378	-0.345	-13.228	16.817	-0.002	-0.002	0.001
45	41	-10.444	-0.220	-13.195	16.829	-0.002	-0.002	0.001
46	23	2.572	-0.210	12.847	13.103	0.002	0.001	0.000
46	19	2.507	-0.116	12.807	13.051	0.001	0.001	0.000
58	42	2.245	-0.104	-12.788	12.984	-0.001	-0.001	0.000
58	23	1.847	-0.248	12.787	12.922	-0.000	0.004	0.000
49	23	-4.437	-0.391	12.781	13.535	0.001	-0.001	0.004
52	23	-6.252	-0.508	12.769	14.227	0.001	-0.001	0.004
58	46	2.356	-0.193	-12.751	12.969	-0.001	-0.001	0.000
58	19	1.736	-0.159	12.750	12.869	-0.000	0.004	0.000
46	42	1.680	-0.162	-12.744	12.855	-0.000	-0.004	0.000
49	19	-5.007	-0.208	12.743	13.693	0.001	-0.001	0.002

	3 DL	0.000	0.000	0.000	0.000	0.000
		1.425	0.000	-0.749	-0.002	0.749
		2.850	0.000	-1.081	-0.002	1.081
		4.275	-0.000	-0.590	-0.001	0.590
		5.700	0.000	0.000	0.000	0.000
	4 LL	0.000	0.000	0.000	0.000	0.000
		1.425	0.000	0.003	-0.000	0.003
		2.850	-0.000	0.003		
		4.275	-0.000	0.002		
		5.700	0.000	0.000		
	5 WL+X	0.000	0.000	0.000	0.000	0.000

Maximum attained Drift as per Analysis from above figure's = **25 mm**

Allowable Drift as per Table-10.12, SBC-301-2007 is H/100 = 6500/100 = **65mm**

Hence the Structure is safe considering drift check…

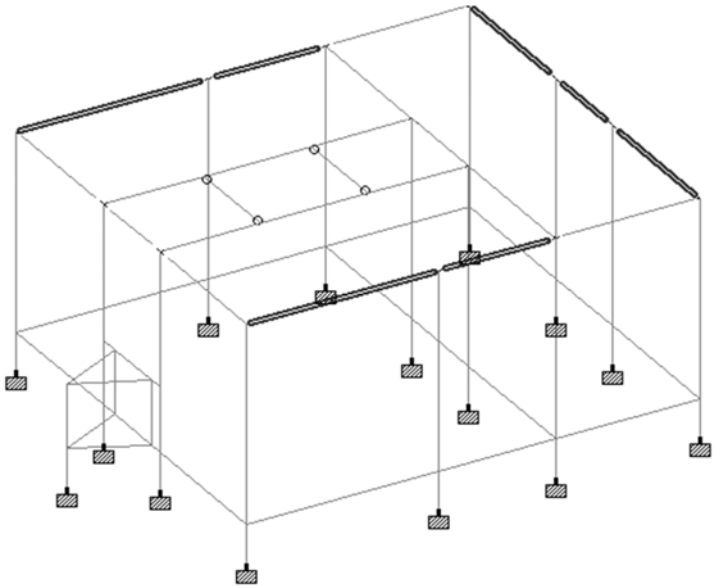
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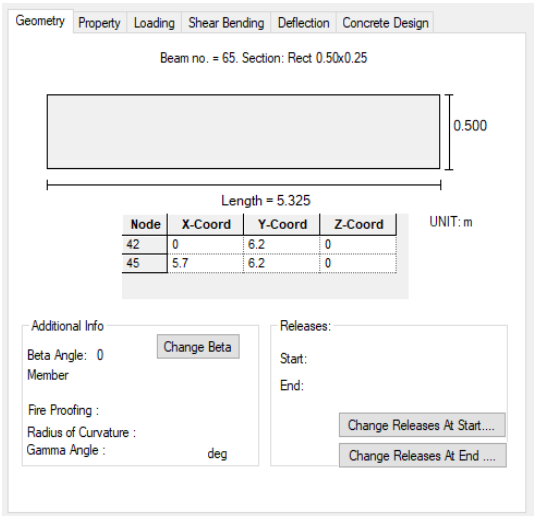
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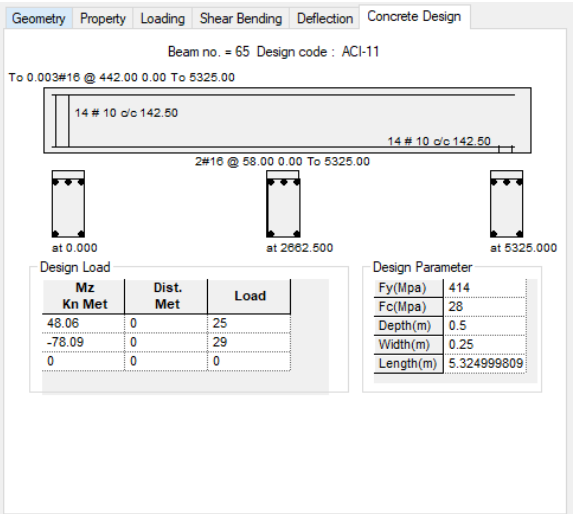
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9. CONCRETE ROOF BEAM DESIGN

9.1. Design of Ledger Beam(LB1)







Design of Ledger Beam (LB1)

Maximum Moment required from Analysis = 78 **Knm (Hogging)**

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Therefore Ast.req = $$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

Ast.req = **480 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f_c'}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **48 Knm (Sagging)**

Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

Ast.pro= **600mm² >>>>>>(Ast.min, Ast.req)**

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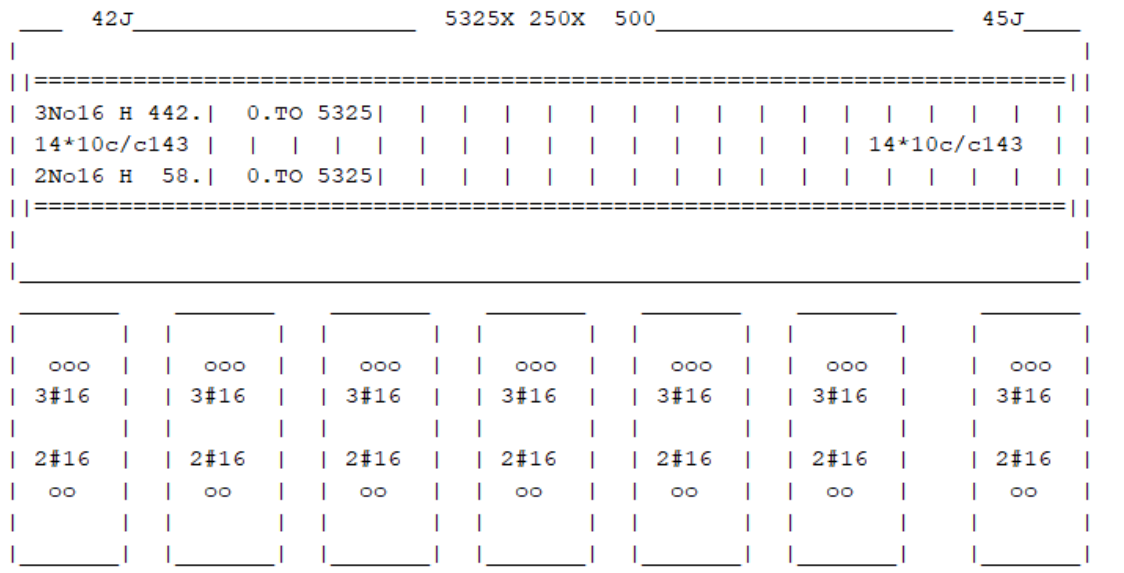
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9.1.1. Staad Output for LB1

```
AT START SUPPORT - Vu=   22.73 KNS  Vc=   89.80 KNS  Vs=    0.00 KNS
Tu=    7.88 KN-MET  Tc=    3.2 KN-MET  Ts=   10.5 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1723. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.60 SQ.CM.

AT END   SUPPORT - Vu=   45.48 KNS  Vc=   93.59 KNS  Vs=    0.00 KNS
Tu=    6.89 KN-MET  Tc=    3.3 KN-MET  Ts=    9.2 KN-MET  LOAD    28
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1723. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   2.27 SQ.CM.
```

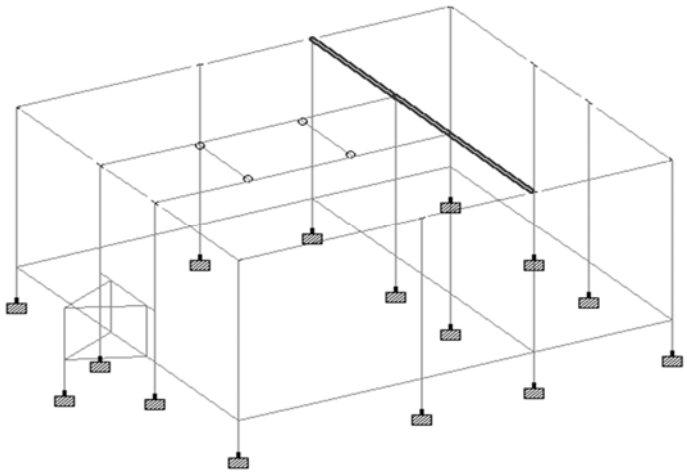


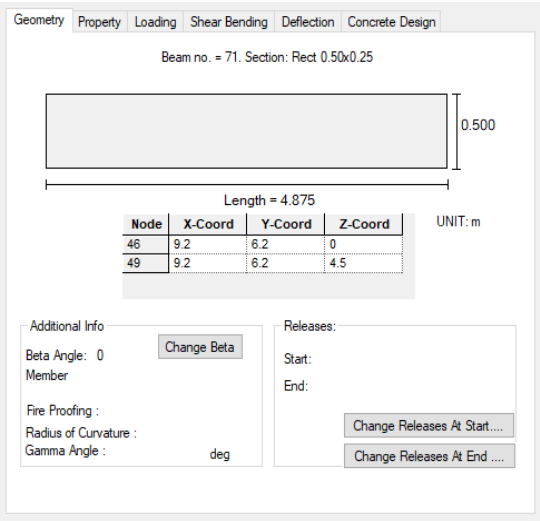
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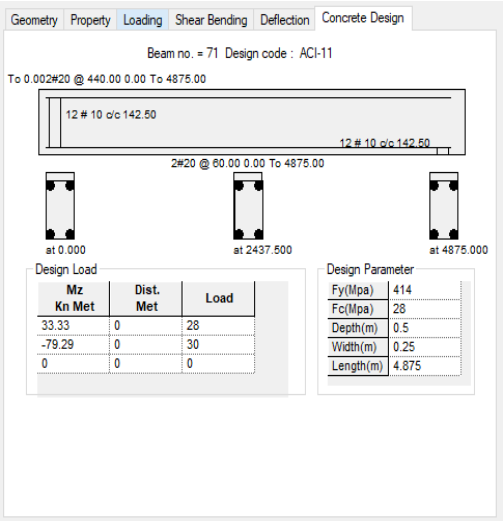
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9.2. Design of Ledger Beam(LB2)







Design of Beam (LB2)

Maximum Moment required from Analysis = 79 **Knm (Hogging)**

Therefore Ast.req =

$$A_s = 0.85 f_c' \frac{b}{Fy}\left(d - \sqrt{d^2 - \frac{2M_u}{0.85\Phi f_c' b}}\right)$$

Where b=250mm, Fy=414 Mpa, d=500-40-10-16/2=442mm, Fc'=35 Mpa & ϕ =0.9

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Ast.req = **485 mm²**

Ast.min= (As per clause-10.5-ACI-318)

$$A_{s,\,min} = \frac{0.25\sqrt{f'_c}}{f_y} b_w d \qquad (10\text{-}3)$$

and not less than $1.4 b_w d / f_y$.

Ast.min= **406mm²**

Maximum Moment required from Analysis = **34 Knm (Sagging)**

Hence Provide 3-20 dia-Full Straight Bars Top & Bottom

Ast.pro= **942mm² >>>>>>(Ast.min, Ast.req)**

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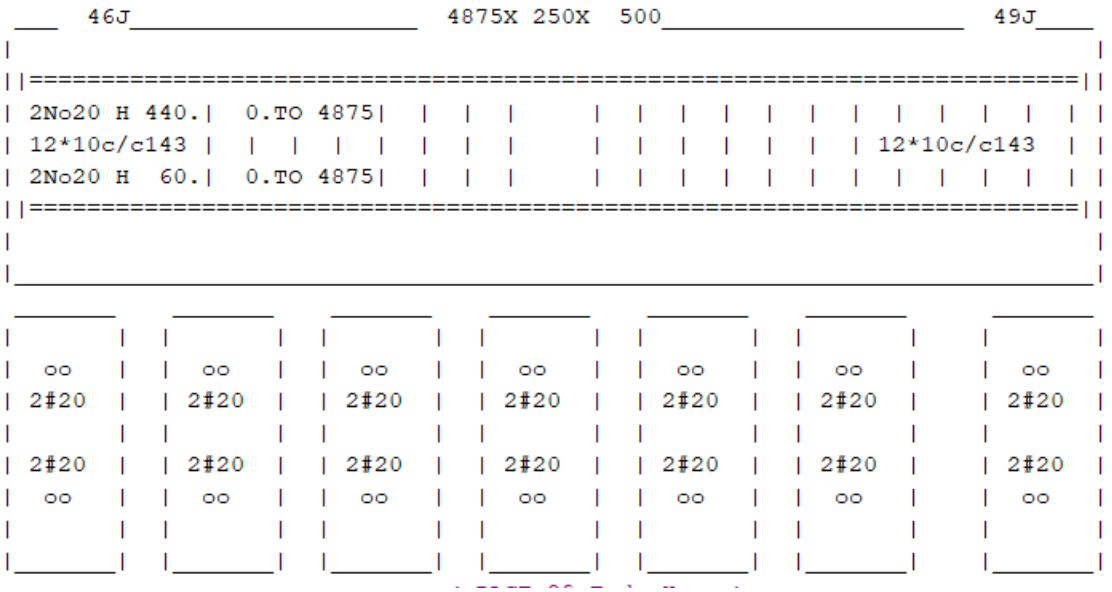
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9.2.1. Staad Output for LB2

```
AT START SUPPORT - Vu=   26.75 KNS  Vc=   95.16 KNS  Vs=    0.00 KNS
Tu=   14.48 KN-MET  Tc=    3.4 KN-MET  Ts=   19.3 KN-MET  LOAD    25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1498. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.77 SQ.CM.

AT END   SUPPORT - Vu=   39.13 KNS  Vc=  113.71 KNS  Vs=    0.00 KNS
Tu=   14.48 KN-MET  Tc=    3.4 KN-MET  Ts=   19.3 KN-MET  LOAD    25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1498. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   4.77 SQ.CM.
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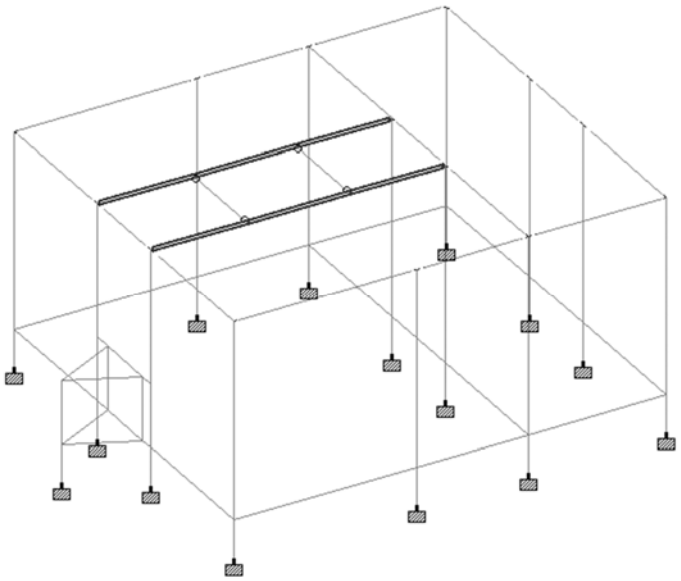


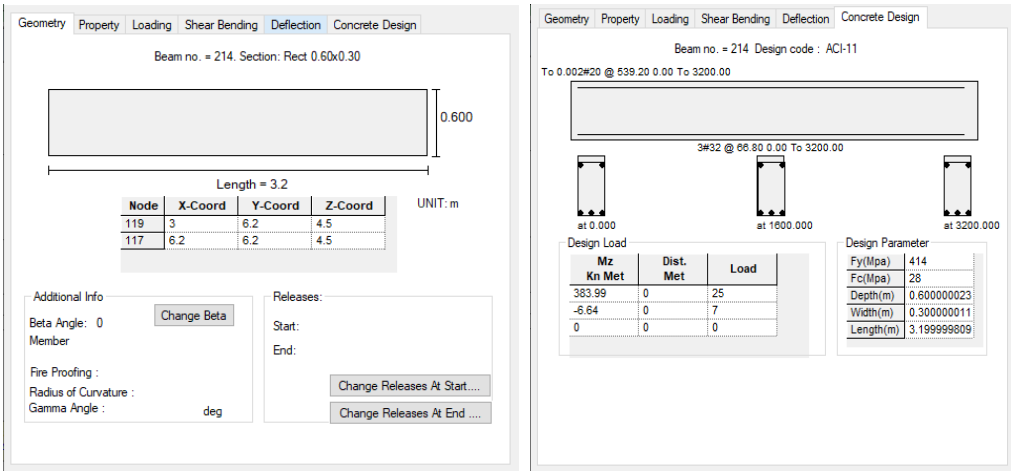
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9.3. Design of Ledger Beam(LB3/LB4)





Design of Beam (LB3)

Provide 6 No's Dia 25@ bottom & 3-Dia 20@ Top considering long term deflection check shown in section 9.3.1

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9.3.1. Longterm Deflection Check for LB3/LB4

INPUT DATA :

Conc. Compressive stress, fc`	=	35.00	MPa
Reinforcement bar tensile stress, fy	=	414.00	MPa
Modulus Elasticity of concrete,	=	27,806	Mpa
Modulus Elasticity of steel, *Es*	=	200,000	Mpa
Unit wt. of concrete, Wconc	=	24.00	kN/m^3
Unit wt. of steel	=	77.00	kN/m^3
Depth of beam, h	=	750	mm
width of beam, b	=	300	mm
Length of beam, L	=	9.00	m
Bottom bar diam., db	=	25.00	mm
area of 1 bar Ab	=	490.87	mm^2
no. of bottom bars	=	6.00	pcs.
Top bar diam., db	=	20.00	mm
area of 1 bar Ab	=	314.16	mm^2
no. of top bars	=	3.00	pcs.
Stirrups diam. Dv	=	10.00	mm
area of 1 bar Av	=	78.54	mm^2
concrete cover, cc	=	40.00	mm
$d_{eff\ bot}$ (@ bottom), d	=	687.50	mm
Mdead	=	268.00	kN-m
M live	=	16.00	kN-m

CHECK DEFLECTION: M_{d+l} = 284.00 kN-m

$$M_{sus} = M_d + 0.50 M_l = 276.00 \text{ kN-m}$$

modulus of rupture, modulus of elasticity, modular ratio

$$f_r = 0.62\lambda\sqrt{f'_c} = 3.67 \text{ Mpa}$$ ←*ACI(318M-08) Sec. 9.5.2.3*

$$n_s = \frac{Es}{Ec} = 7.19$$

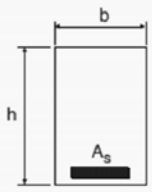
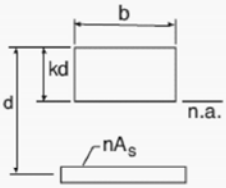
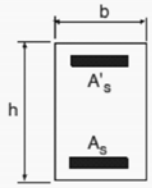
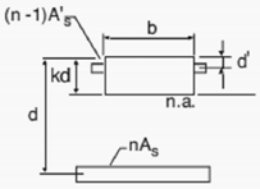
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gross and crack section of moment of inertia

Table 10-2 Gross and Cracked Moment of Inertia of Rectangular and Flanged Section

Gross Section	Cracked Transformed Section	Gross and Cracked Moment of Inertia
	 Without compression steel	$n = \frac{E_s}{E_c}$ $B = \frac{b}{(nA_s)}$ $I_g = \frac{bh^3}{12}$ Without compression steel $kd = \left(\sqrt{2dB+1}-1\right)/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2$
	 With compression steel	With compression steel $r = (n-1)A'_s/(nA_s)$ $kd = \left[\sqrt{2dB(1+rd'/d)+(1+r)^2}-(1+r)\right]/B$ $I_{cr} = b(kd)^3/3 + nA_s(d-kd)^2 + (n-1)A'_s(kd-d')^2$

$$I_g = \frac{bh^3}{12} = 1.05E+10 \text{ mm?}$$

$$B = \frac{b}{nAs} = 0.01 \text{ mm/mm²}$$

$$kd = \left(\sqrt{2dB+1}-1\right)/B = 248.89 \text{ mm}$$

$$I_{cr} = \frac{b(kd)^3}{3} + nAs(d-kd)^2 = 5.62E+09 \text{ mm?}$$

$$\frac{I_g}{I_{cr}} = 1.88$$

effective moment of inertia

$$M_{cr} = \frac{frI_g}{y_t} = 103.16 \text{ kN-m} \qquad \text{?ACI(318M-08) Sec. 9.5.2.3}$$

under dead load

$$\frac{M_{cr}}{M_d} = 0.38$$

$$(I_e)_d = \left(\frac{M_{cr}}{M_d}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_d}\right)^3\right] I_{cr} = 5.90E+09 \text{ mm?} \qquad \text{?ACI(318M-08) Sec. 9.5.2.3}$$

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under sustained load

$$\frac{M_{cr}}{M_{sus}} = 0.37$$

$$(I_e)_{sus} = \left(\frac{M_{cr}}{M_{sus}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{sus}}\right)^3\right] I_{cr} = 5.87E+09 \text{ mm?}$$ *?ACI(318M-08) Sec. 9.5.2.3*

under dead + live load

$$\frac{M_{cr}}{M_{d+\ell}} = 0.36$$

$$(I_e)_{d+\ell} = \left(\frac{M_{cr}}{M_{d+\ell}}\right)^3 I_g + \left[1-\left(\frac{M_{cr}}{M_{d+\ell}}\right)^3\right] I_{cr} = 5.85E+09 \text{ mm?}$$ *?ACI(318M-08) Sec. 9.5.2.3*

initial or short-term deflection

$$K = 1.00$$

$$(\Delta_i)_d = \frac{K(5/48)M_d\ell^2}{Ec(I_e)_d} = 13.787 \text{ mm}$$

$$(\Delta_i)_{sus} = \frac{K(5/48)M_{sus}\ell^2}{Ec(I_e)_{sus}} = 14.256 \text{ mm}$$

$$(\Delta_i)_{d+\ell} = \frac{K(5/48)M_{d+\ell}\ell^2}{Ec(I_e)_{d+\ell}} = 14.723 \text{ mm}$$

$$(\Delta_i)_\ell = (\Delta_i)_{d+\ell} - (\Delta_i)_d = 0.935 \text{ mm}$$

Table 10-3 Deflection Coeffcient K

		K
1.	Cantilevers (deflection due to rotation at supports not included)	2.40
2.	Simple beams	1.0
3.	Continuous beams	1.2-0.2 M_o/M_a
4.	Fixed-hinged beams (midspan deflection)	0.80
5.	Fixed-hinged beams (maximum deflection using maximum moment)	0.74
6.	Fixed-fixed beams	0.60

For other types of loading, K values are given in Ref. 10.2.

M_o = Simple span moment at midspan $\left(\frac{w\ell^2}{8}\right)$

M_a = Net midspan moment.

TABLE 9.5(b) — MAXIMUM PERMISSIBLE COMPUTED DEFLECTIONS

Type of member	Deflection to be considered	Deflection limitation
Flat roofs not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/180^*$
Floors not supporting or attached to nonstructural elements likely to be damaged by large deflections	Immediate deflection due to live load *L*	$\ell/360$
Roof or floor construction supporting or attached to nonstructural elements likely to be damaged by large deflections	That part of the total deflection occurring after attachment of nonstructural elements (sum of the long-term deflection due to all sustained loads and the immediate deflection due to any additional live load)†	$\ell/480^‡$
Roof or floor construction supporting or attached to nonstructural elements not likely to be damaged by large deflections		$\ell/240^§$

*Limit not intended to safeguard against ponding. Ponding should be checked by suitable calculations of deflection, including added deflections due to ponded water, and considering long-term effects of all sustained loads, camber, construction tolerances, and reliability of provisions for drainage.

†Long-term deflection shall be determined in accordance with 9.5.2.5 or 9.5.4.3, but may be reduced by amount of deflection calculated to occur before attachment of nonstructural elements. This amount shall be determined on basis of accepted engineering data relating to time-deflection characteristics of members similar to those being considered.

‡Limit may be exceeded if adequate measures are taken to prevent damage to supported or attached elements.

§Limit shall not be greater than tolerance provided for nonstructural elements. Limit may be exceeded if camber is provided so that total deflection minus camber does not exceed limit.

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$$(\Delta_i)_\ell \le \frac{\ell}{180} = 50.00 \text{ mm} > 0.94 \text{ mm} \quad ok!$$

$$(\Delta_i)_\ell \le \frac{\ell}{360} = 25.00 \text{ mm} > 0.94 \text{ mm} \quad ok!$$

additional long-term deflection

$$\xi = 2.00$$

5 years or more	2.0
12 months	1.4
6 months	1.2
3 months	1.0

$$\rho = \frac{As}{bd} = 0.0143$$

$$\lambda_\Delta = \frac{\xi}{1+50\rho'} = 1.17 \quad ?ACI(318M\text{-}08)\ Sec.\ 9.5.2.5$$

$$\Delta_{cp} + \Delta_{sh} = \lambda(\Delta_i)_{sus} = 16.64 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell = 17.57 \text{ mm}$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{480} = 18.75 \text{ mm} > 17.57 \text{ mm} \quad ok!$$

$$\Delta_{cp} + \Delta_{sh} + (\Delta_i)_\ell \le \frac{\ell}{240} = 37.50 \text{ mm} > 17.57 \text{ mm} \quad ok!$$

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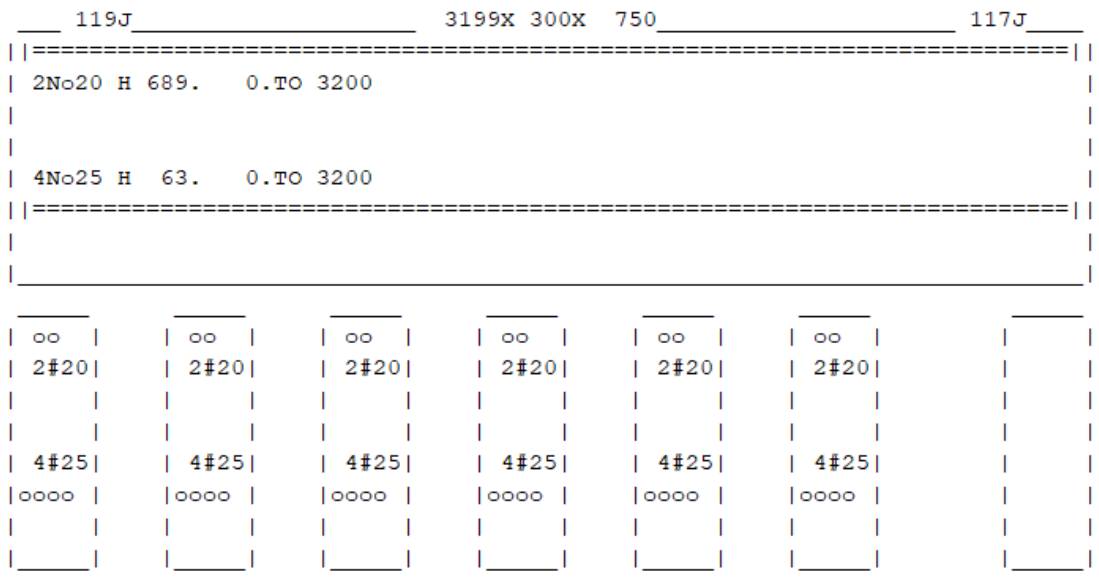
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9.3.2. Staad Output for LB3/LB4

```
                B E A M  N O.   214 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=   13.29 KNS  Vc=  171.83 KNS  Vs=    0.00 KNS
  Tu=     0.52 KN-MET  Tc=    8.0 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                     STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   13.27 KNS  Vc=  171.83 KNS  Vs=    0.00 KNS
  Tu=     0.52 KN-MET  Tc=    8.0 KN-MET  Ts=    0.0 KN-MET  LOAD     3
                     STIRRUPS ARE NOT REQUIRED.
```

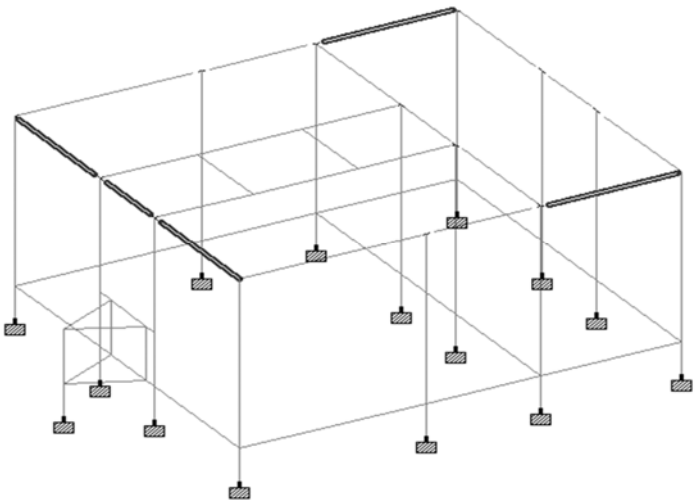


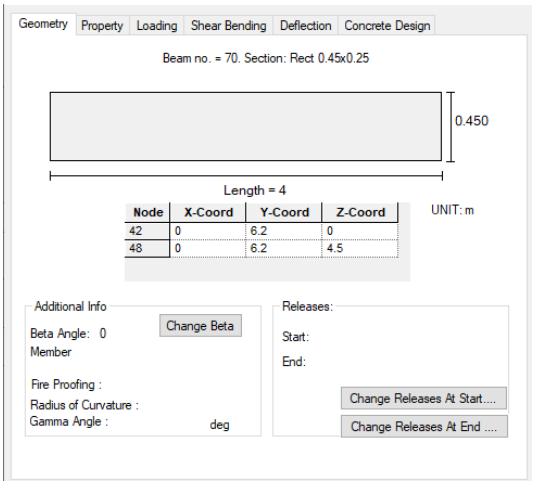
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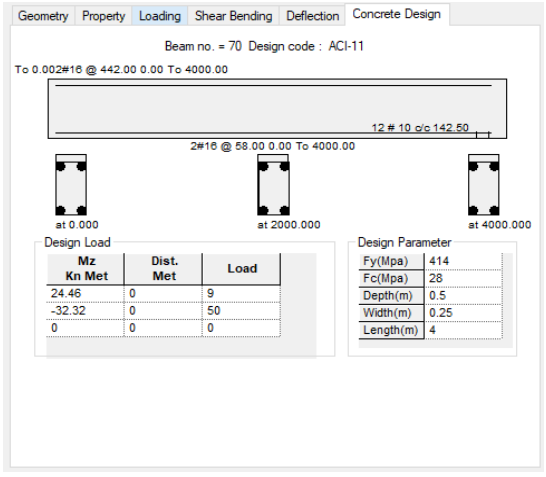
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9.4. Design of Beam(B1)







Hence Provide 3-16 dia-Full Straight Bars Top & Bottom

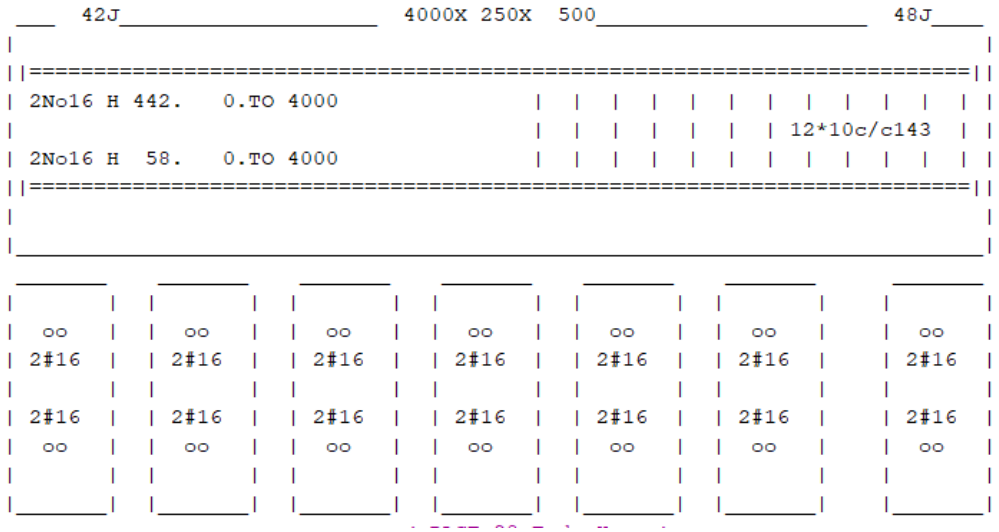
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9.4.1. Staad Output for B1

```
AT START SUPPORT - Vu=     3.77 KNS   Vc=  169.04 KNS   Vs=     0.00 KNS
Tu=     1.41 KN-MET   Tc=     3.4 KN-MET   Ts=     0.0 KN-MET   LOAD      3
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=     5.50 KNS   Vc=  115.40 KNS   Vs=     0.00 KNS
Tu=     5.84 KN-MET   Tc=     3.4 KN-MET   Ts=     7.8 KN-MET   LOAD     25
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM  C/C FOR 1560. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =    1.92 SQ.CM.
```

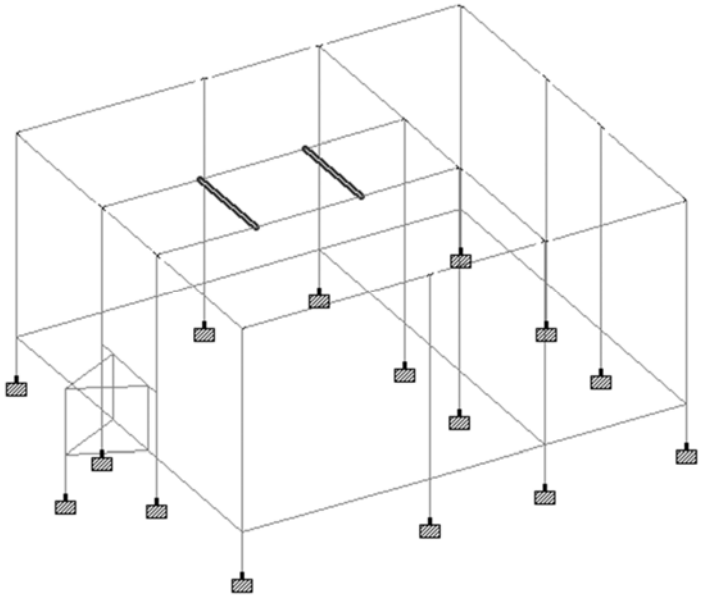


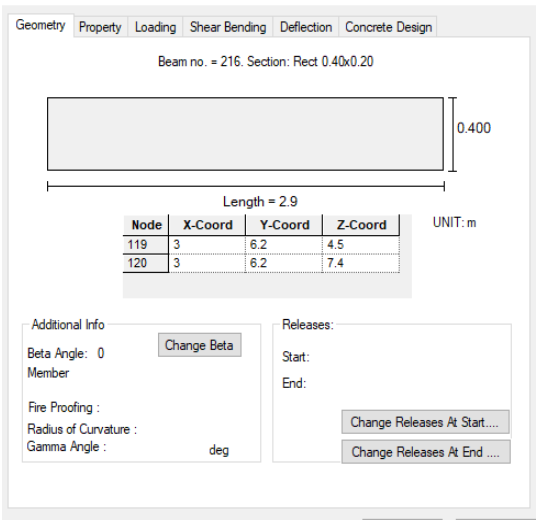
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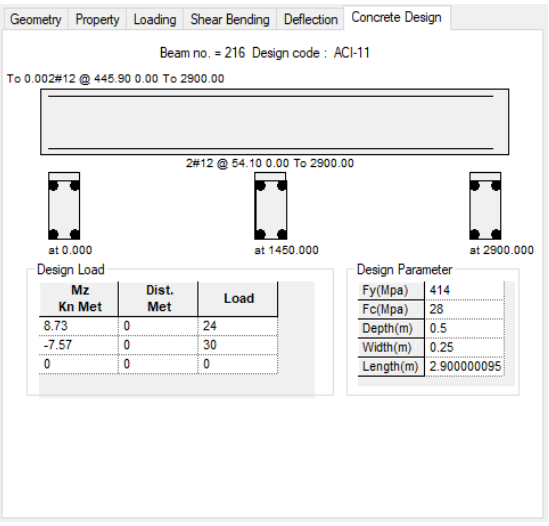
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9.5. Design of Beam-(B2)







However Provide minimum 3-20 dia-Full Straight Bars Top & Bottom considering hoop stresses from Ring beam.

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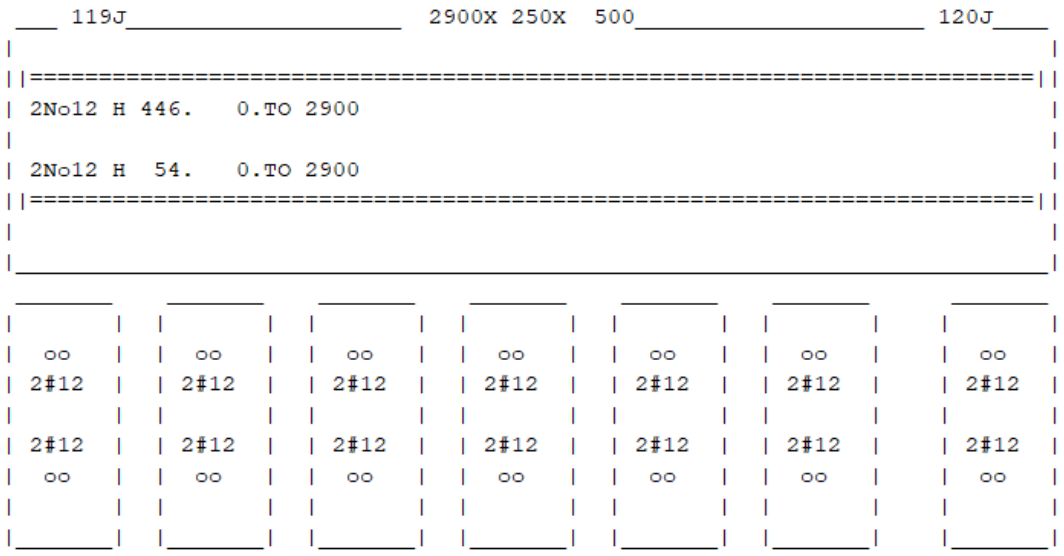
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9.5.1. Staad Output for B2

```
        B E A M  N O.   216 D E S I G N  R E S U L T S - SHEAR


  AT START SUPPORT - Vu=     3.98 KNS   Vc=    96.99 KNS   Vs=     0.00 KNS
  Tu=     0.01 KN-MET   Tc=     3.4 KN-MET   Ts=     0.0 KN-MET   LOAD     3
                     STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=     4.10 KNS   Vc=    96.99 KNS   Vs=     0.00 KNS
  Tu=     0.01 KN-MET   Tc=     3.4 KN-MET   Ts=     0.0 KN-MET   LOAD     3
                     STIRRUPS ARE NOT REQUIRED.
```

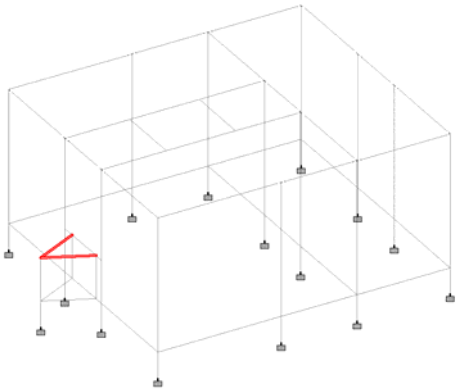


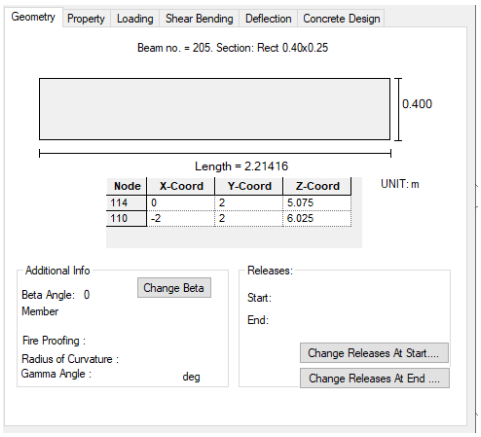
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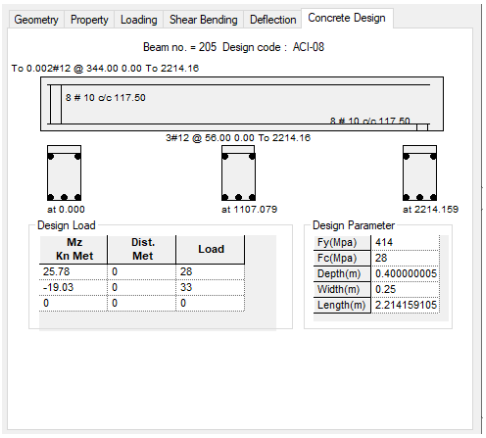
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9.6. Design of Beam-(B4)







However Provide minimum 3-12 dia-Full Straight Bars Top & Bottom.

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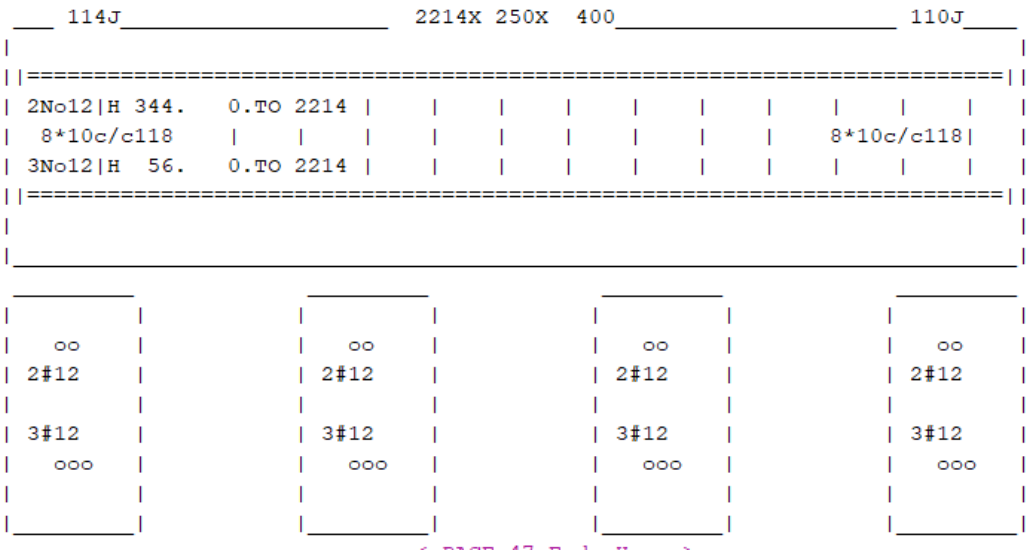
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9.6.1. Staad Output for B4

```
        B E A M  N O.   205 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=    2.03 KNS  Vc=   71.46 KNS  Vs=    0.00 KNS
Tu=    5.08 KN-MET  Tc=    2.6 KN-MET  Ts=    6.8 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 118. MM  C/C FOR  767. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.83 SQ.CM.

AT END   SUPPORT - Vu=   15.59 KNS  Vc=   81.15 KNS  Vs=    0.00 KNS
Tu=    5.08 KN-MET  Tc=    2.6 KN-MET  Ts=    6.8 KN-MET  LOAD    48
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 118. MM  C/C FOR  767. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE =   1.83 SQ.CM.
```



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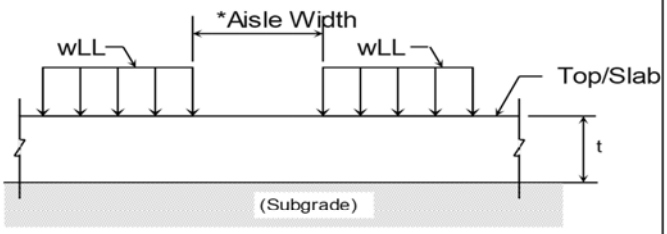
10. CONCRETE GRADE SLAB DESIGN

CONCRETE SLAB ON GRADE ANALYSIS

For Slab Subjected to Stationary Uniformly Distributed Live Loads

Input Data:

Slab Thickness, t =	3.937	in.	100.000	mm		
Concrete Strength, f 'c =	5000	psi	35.000	kpa		
Subgrade Modulus, k =	39.7866	pci	10800	Kn/m3	(AS PER SECTION-7.7-SOIL REPORT)	
Factor of Safety, FS =	1.500		1.500			
Uniform Live Load, wLL =	104.45	psf	5.00	kpa		



Concrete Slab on Grade with Uniform Loads

Results:

Design Parameters:

Modulus of Rupture, MR =	636.40	psi	4387.79	kpa	MR = 9*SQRT(f 'c)
Allow. Bending Stress, Fb =	424.26	psi	2925.19	kpa	Fb = MR/FS
Modulus of Elasticity, Ec =	4030509	psi	27789329	kpa	Ec = 57000*SQRT(f 'c)
Poisson's Ratio, μ =	0.15		0.15		μ = 0.15 (assumed for concrete)
Radius of Stiffness, Lr =	26.94	in.	684.37	mm	Lr = (Ec*t^3/(12*(1-μ^2)*k))^0.25
Critical Aisle Width, Wcr =	4.96	ft.	1511.77	mm	Wcr = (2.209*Lr)/12

Stationary Uniformly Distributed Live Loads:

wLL(allow) =	682.05	psf	wLL(allow) = 257.876*Fb*SQRT(k*t/Ec)
	32.65	kpa	**wLL(allow) >= wLL, O.K.**

Provide Minimum Reinforcement of 0.14%@both faces = 0.14%(h) = 140 mm2/M

Table 24.4.3.2—Minimum ratios of deformed shrinkage and temperature reinforcement area to gross concrete area

Reinforcement type	f_y, psi	Minimum reinforcement ratio	
Deformed bars	< 60,000	0.0020	
Deformed bars or welded wire reinforcement	≥ 60,000	Greater of:	$\frac{0.0018 \times 60,000}{f_y}$
			0.0014

Provide WWF-8mm Dia@200 c-c Both Ways-TOP LAYER

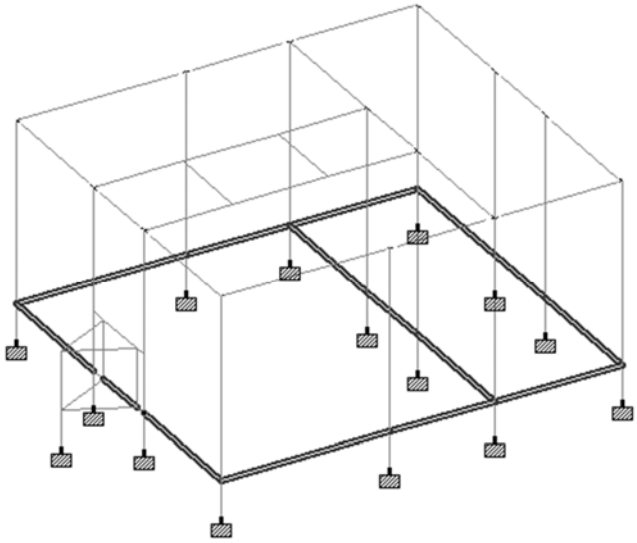
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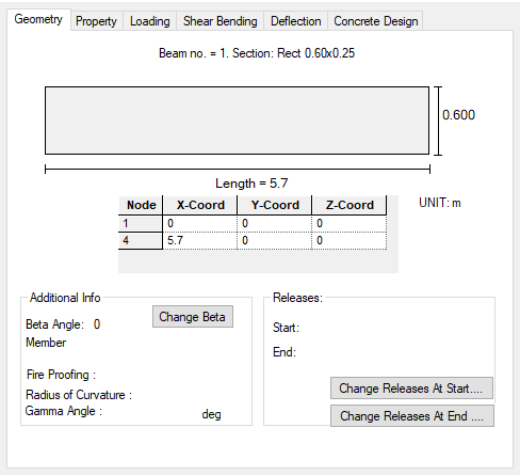
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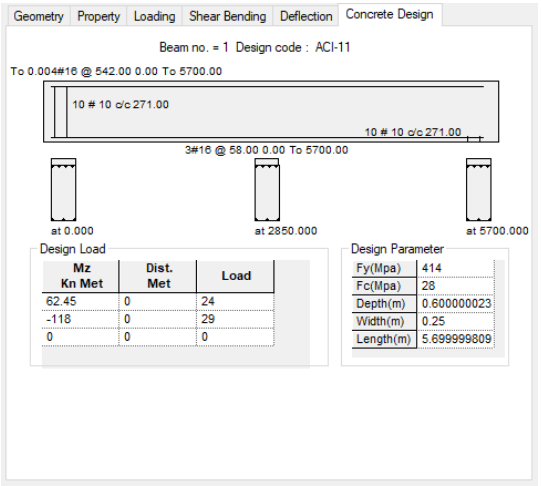
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11. DESIGN OF GRADE BEAM

11.1. Grade Beam(GB1)







Hence Provide 3-20 dia-Full Straight Bars Top & 3-20 dia-Full Straight Bars Bottom

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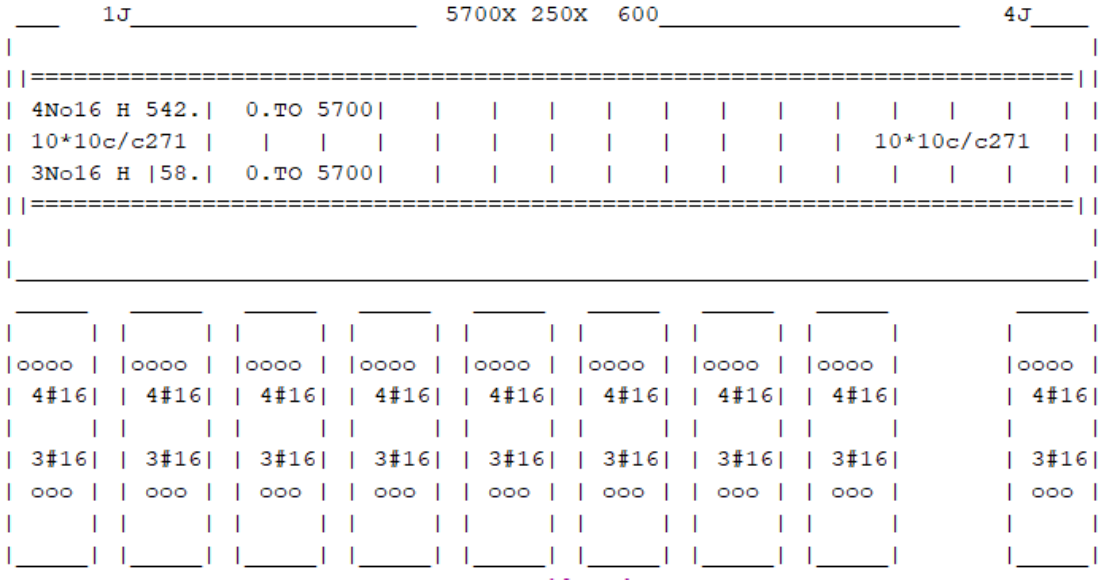
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11.1.1. Staad Output for GB1

```
        B E A M  N O.     1 D E S I G N  R E S U L T S - SHEAR


AT START SUPPORT - Vu=   56.43 KNS  Vc=  144.33 KNS  Vs=    0.00 KNS
Tu=    0.25 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2310. MM

AT END   SUPPORT - Vu=   67.12 KNS  Vc=  120.60 KNS  Vs=    0.00 KNS
Tu=    0.25 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD     3
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 271. MM  C/C FOR 2310. MM
```



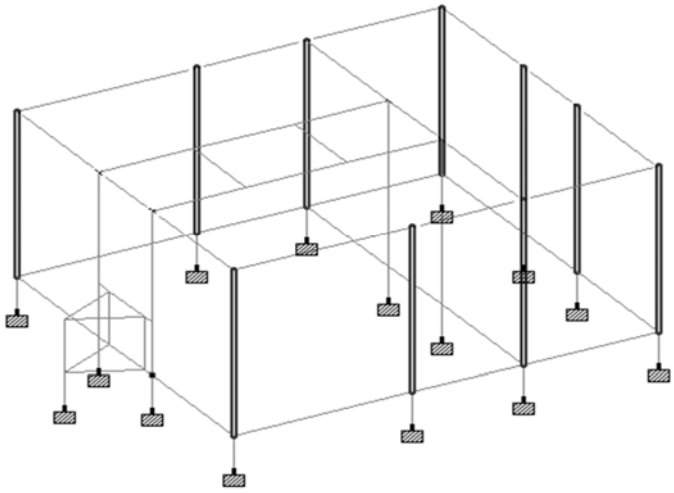
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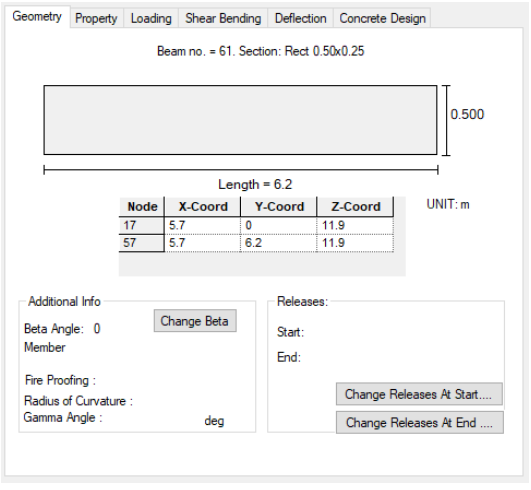
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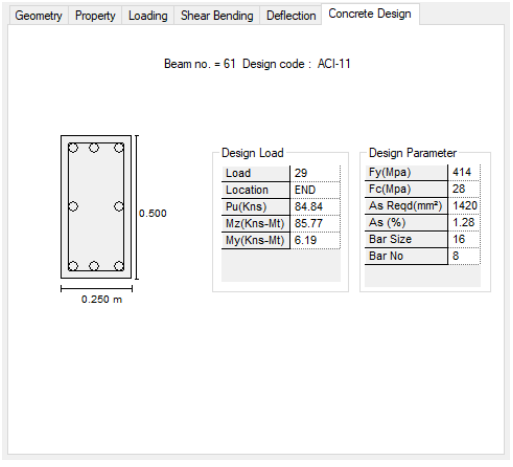
12. CONCRETE COLUMN DESIGN

12.1. Design of Column-(C1)



Column C1





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12.1.1. Staad Output for Column C1

```
ACI 318-11   COLUMN  NO.    61   DESIGN RESULTS
=================================================


FY - 413.7  FC -  27.6 MPA,  RECT SIZE - 250.0 X 500.0 MMS, TIED
        AREA OF STEEL REQUIRED =  1420.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-----------------------------------------------------------

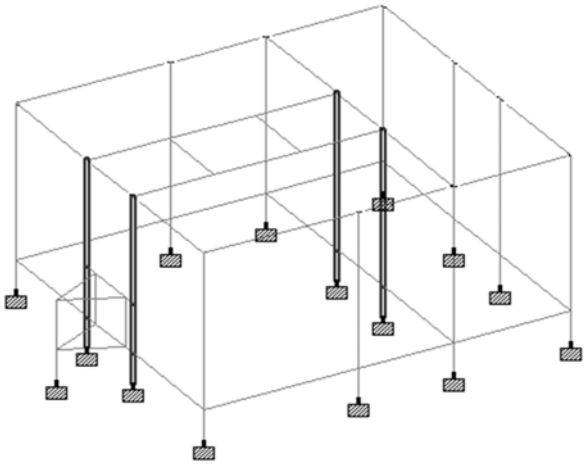
  8 - 16 MM               1.287       29      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 250.00 MM
```


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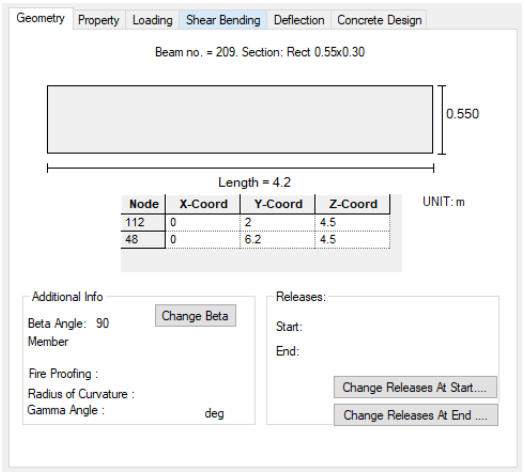
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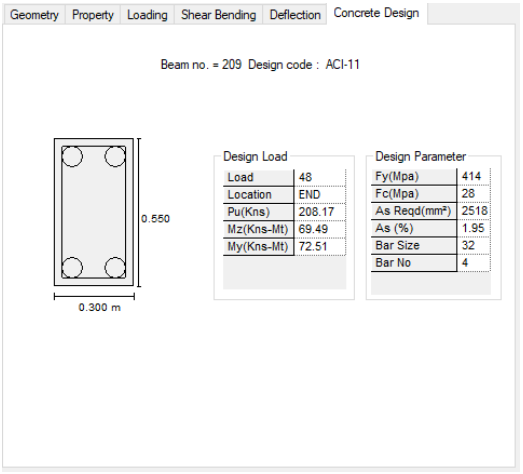
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12.2. Design of Column-(C2)



Column numbers C2





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12.2.1. Staad Output for Column C2

```
        ACI 318-11   COLUMN  NO.   209   DESIGN RESULTS
        ==============================================


FY - 413.7   FC -   27.6 MPA,   RECT SIZE - 300.0 X 550.0 MMS, TIED
          AREA OF STEEL REQUIRED =   2517.9   SQ. MM

BAR CONFIGURATION        REINF PCT.    LOAD    LOCATION    PHI
-------------------------------------------------------------

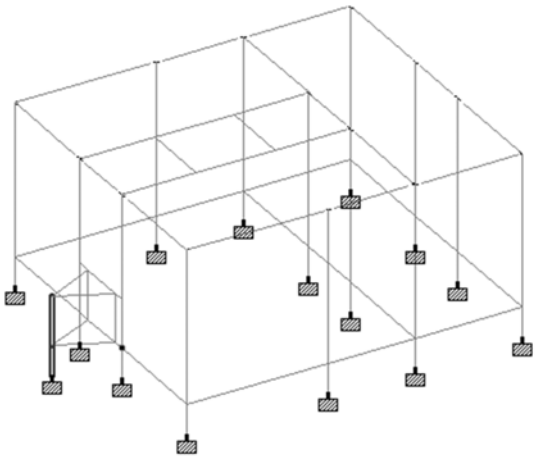
  4 - 32 MM                  1.950        48       END      0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 192.00 MM
```


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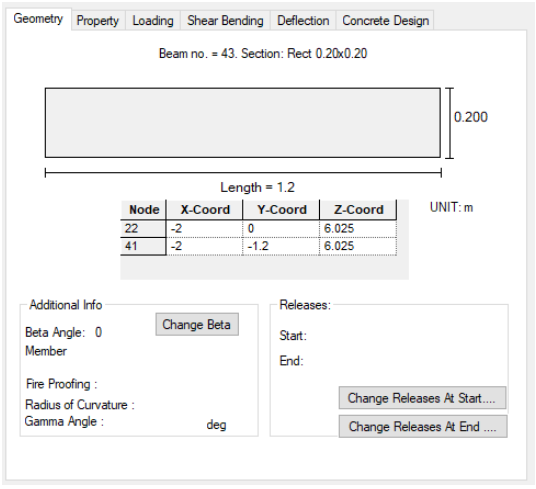
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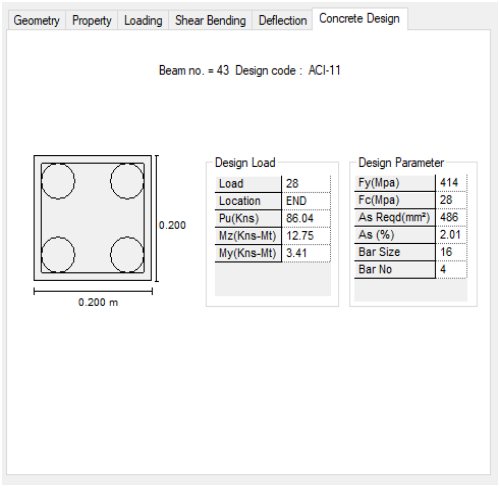
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12.3. Design of Column-(C3)



Column numbers C3





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12.3.1. Staad Output for Column C3

```
      ACI 318-11   COLUMN  NO.    43   DESIGN RESULTS
      ================================================


FY - 413.7  FC -  27.6 MPA,  SQRE SIZE - 200.0 X 200.0 MMS, TIED
        AREA OF STEEL REQUIRED =   485.6  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
-------------------------------------------------------------

  4 - 16 MM               2.011       28      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 192.00 MM
```


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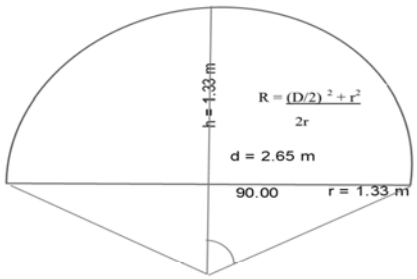
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13. DOME & RING BEAM(B3) DESIGN

Dome & Ring Beam Design as per ACI-318-08(PCA Notes)

Dimensions of Dome

Diameter	d =	2650 mm
Height	h =	1325 mm
Thickness	t =	120 mm
Radius of Sphere	R =	1325 mm
	Φ =	90.00
	Θ =	0 to 90.00



Loading

Dead Load	DL =	2.88 KN/m
Live Load	LL =	0.10 KN/m
Wind Load	WL =	0.10 KN/m
Total Load	W =	3.08 KN/m
Factored Load	Wu =	4.62 KN/m

Meridional Stress =WR/t(1/1+cosθ)
Hoop Stresses =WR/tX((cos^2(θ)+cosθ-1)/(1+cosθ))

Meridional Stress

Θ	Mt
90.00	0.051 MPa
45.00	0.030 MPa
40.00	0.029 MPa
35.00	0.028 MPa
30.00	0.027 MPa
25.00	0.027 MPa
20.00	0.026 MPa
15.00	0.026 MPa
5.00	0.026 MPa
0.00	0.026 MPa

Maximum Meridional Stress 0.051 MPa

Hoop Stress

Θ	Mt
90.00	-0.032 MPa
45.00	0.004 MPa
40.00	0.006 MPa
35.00	0.009 MPa
30.00	0.010 MPa
25.00	0.012 MPa
20.00	0.013 MPa
15.00	0.015 MPa
5.00	0.016 MPa
0.00	0.016 MPa

Maximum Hoop Stress 0.032 MPa

fck	35
Fy	415 MPa
6st	230.000 MPa

Area of steel 27 sqmm

Bar Dia 10 mm
Spacing **2951 c/c**

Area of steel 17 sqmm

Bar Dia 10 mm
Spacing **4734 c/c**

However Provide Minimum Reinf both Circumferentially & meridionally in the Dome as per Clause 19.4.9

Astmin= 0.35%*t*h = 557 sqmm

Spacing of Reinf. 1000*(Abar)/Astmin
= 140 mm
Both Ways

19.4.8 Concentration of Reinforcement

In the case of long barrel shells (or domes) it is often desirable to concentrate tensile reinforcement near the edges rather than distribute the reinforcement over the entire tensile zone. When this is done, a minimum amount of reinforcement equal to 0.0035bh must be distributed over the remaining portion of the tensile zone, as shown in Fig. 27-3. This amount in practical terms is twice the minimum steel requirement for shrinkage and temperature stresses.

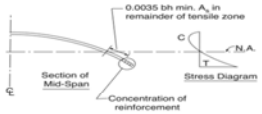


Figure 27-3 Concentration of Shell Reinforcement

19.4.10 Spacing of Reinforcement

Maximum permissible spacing of reinforcement is the smaller of 5 times the shell thickness and 18 in. Therefore, for shells less than 3.6 in. thick, 5 times the thickness controls. For thicker shells, the spacing of bars must not exceed 18 in.

Ring Beam Design

The Meredian thrust@support will have a horizontal component which will cause supporting walls to burst outwards, causing its failure. In order to bear this horizontal component, a Ring Beam is provided @Base of Dome.

Meridional Thrust @ Base(T1) **11 KN/m** Wr/(1+cos Θ)

Hoop Tension on Ring Beam(Th) = **6 KN**
Area of steel 15 sqmm T1*(cos Θ)* D/2

However Provide minimum 3-16mm Dia T&B-Full Straight Bars

Bar Dia 16 mm
No of Bars **3 nos**

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14. CANTILEVER SLAB DESIGN

DESIGN OF CANTILEVER SLAB

Geometrical Properties			Material Properties		
Span to center of support "sp_1"	0.8	met	Weight of reinforced concrete "γ_r	24	kN/m^3
Clear Span	0.6	met	Strength of concrete "f'_c"	35	N/mm^2
Clear span + depth of slab "sp_2"	0.75	met	Strength of steel "f_y"	420	N/mm^2
Max. eff span le = min (sp_1;sp_2)	0.8	met	Clear cover to concrete("c")	20	mm
Width of Slab "b"	1000	mm	Flexure strength reduction factor	0.9	
Thickness of slab "t"	150	mm	Shear strength reduction factor	0.75	
Eff. Depth "d"	125		Modulus of Elasticity of steel "Es"	2E+05	N/mm^2
Loading			Modulus of Elasticity of concrete	27806	N/mm^2
Self Weight of slab	3.6	kN/m^2	Modular Ratio "n = Es/Ec"	7.19	
Floor Finish	3.5	kN/m^2	**Provided Reinforcement**		
Wall load	0	kN/m^2	Top layer Nos.	6 Φ 10	mm
Total dead Load "U_{dl}"	7.1	kN/m^2	Bottom layer Nos.	0 Φ 12	mm
Live load	1	kN/m^2	Area of Top steel "$A_{s\ t}$"	468	mm^2
Total Ultimate Load "$Uu_{,tot}$"$_{(1.2D+1.6L)}$	10.12	kN/m^2	Area of Bottom steel "$A_{s\ b}$"	0	mm^2
Moment due to Overhang Point Load(0.125*0.15*24*0.6)	0.27	kNm	"d_b' = 10 mm		
Calculations					
Shear force Vu = U*le	**8.096**	**kN**			
Neg. Moment Mu = U*le^2/2	**3.51**	**kNm**			

Check for Section capacity								
Depth of comp. block "a" = $A_{s\ t}$*f_y/(0.85*f'_c*b)	6.61	mm						
Section capacity ΦMn = Φ*$A_{s\ t}$*f_y*(d-a/2)	**21.94**	**kNm**	**O.K**	>	**3.51**	**kNm**	(Mu)	
Shear requirements								
Shear capacity ΦVc = Φ*0.17*√f'c*b*d	**94.29**	**kN**	**O.K**	>	**8.096**	**kN**	(Vu)	
Serviceability requirements								
d' = c+d_b/2	26	mm						
B = b/(n*$A_{s\ t}$)	0.297							
r = (n-1)*$A_{s\ b}$/(n*$A_{s\ t}$)	0							
k_d = (√(2*d*B(1+r*d'/d)+(1+r)^2)-(1+r))/B	29.01	mm						
I_{cr} = bkd^3/3+n*$A_{s\ t}$*(d-k_d)^2+(n-1)*$A_{s\ b}$*(k_d-d'^{2}	3.915E+07	mm^4						
Calculate Gross moment of inertia								
Area of slab A = t*b	2E+05	mm^2						
Depth of neutral axis "y"	75	mm						
Distance of top fiber from N.A "y_t"	75	mm						
Gross moment of inertia "I_g"	2.813E+08	mm^4						
Allowable stress fr = 0.62√f'c	3.67	mm^2						
Cracking moment M_{cr} = fr*I_g/y_t	13.75	kNm						

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Service moments

Moment due to dead load "M_{dl}"	2.54	kNm
Moment due to live load "Mll"	0.32	kNm
Total Moment Ms = $M_{dl}+M_{ll}$	2.86	kNm

Deflection due to dead + live loads

R(dl+ll) = M_{cr}/Ms	1.000		
$I_e(dl+ll) = R^3*I_g+(1-R^3)*I_{cr}$	2.813E+08	mm^4	
$\delta = K*5/48*Ms*le^2/(Ec*I_e)$	0.08	mm	**K = 3.2 (conservative)**

Deflection due to dead load

R(dl) = M_{cr}/Md	5.411	
$I_e(dl) = R^3*I_g+(1-R^3)*I_{cr}$	3.839E+10	mm^4
$\delta = K*5/48*Md*le^2/(Ec*I_e)$	0.00	mm

Deflection due to live load

Live load deflection = dl+ll deflection - dl deflection 0.08 mm

Allowable immediate deflection due to Live load =
per ACI 318 Tab 9.5b $I_e/360$ **2.22 mm > 0.08 mm O.K**

Long term deflection

% of liveload sustained "SL"	50	%
$M_{sus} = M_{dl}+SL*M_{ll}$	2.70	kNm
R(sus) = $M_{cr}/Msus$	5.091	
$I_e(sus) = R^3*I_g+(1-R^3)*I_{cr}$	3.198E+10	
$\delta = K*5/48*Msus*le^2/(Ec*I_e)$	0.00	mm

Magnification factor for creep and shrinkage

$\lambda_\Delta = \xi / (1+50*\rho')$ 2.000

Where:-

ξ = 2 for 5 years or more cl.9.5.2.5 ACI 318

$\rho' = A_{s\ b}/(b*d)$ 0

Total deflection = λ_Δ*sus deflection + ll deflection 0.079 mm

Allowable deflection = $I_e/240$ 3.333 mm > 0.079 mm O.K

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15. FOUNDATION DESIGN

Foundations have been designed based on the Soil Report Recommendations. The minimum Embedment Depth of Foundations has been considered as 1.2m & applicable SBC of 350 Kpa. (Refer Doc-MD-216-3000-EG-CV-RPT-1008 for Details)

TABLE 4: RECOMMENDED NET ALLOWABLE BEARING CAPACITIES

PRPJECT: MANSOURAH - MANSSARAH GOLD PROJECT
CLIENT: L & T CONSTRUCTION

Profile / Borehole	Foundation Type	Depth of Foundation below FGL	Footing Size (B x L)	Net Allowable Bearing Capacity Based on		Recommended Net Allowable Bearing Capacity	Estimated Total Settlement
				Settlement	Shear Failure		
-	-	(m)	(m)	(kN/m^2)	(kN/m^2)	(kN/m^2)	(mm)
BH-01	Square Isolated	1.20	1.0 x 1.0	1500	570	570	25
		1.50		2000	850	850	25
		1.20	2.0 x 2.0	1000	750	750	25
		1.50		1200	980	980	25
		1.20	3.0 x 3.0	775	900	775	25
		1.50		950	1120	950	25
		1.20	4.0 x 4.0	625	1040	625	25
		1.50		750	1250	750	25
		1.20	5.0 x 5.0	525	1172	525	25
		1.50		600	1342	600	25

Refer Attachment- II for Foundation Design calculations

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REFERENCES

1.)	BUILDING PLANS	MD-216-1800-EG-AR-GAD-3165
2.)	OVERALL LAYOUT PLAN	MD-216-1800-EG-GAD-1535
3.)	LAYOUT & DETAILS OF FDN, COL &GRADE BEAM	MD-216-1800-EG-CV-GAD-1548
4.)	LAYOUT & DETAILS OF ROOF & GRADE SLAB	MD-216-1800-EG-CV-GAD-1570
5.)	DETAILS OF ROOF DOME & WALL SECTIONS	MD-216-1800-EG-CV-GAD-1571
6.)	GEOTECHNICAL REPORT	MD-216-3000-EG-CV-RPT-1008
7.)	STRUCTURAL DESIGN CRITERIA	MD-216-0000-EG-ST-CRT-0001
8.)	BUILDING ELEVATIONS & SECTION	MD-216-1800-EG-AR-GAD-3166

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ATTACHMENT 1 STAAD INPUT DATA

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STAAD INPUT FOR MOSQUE:

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 18-Oct-19
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 0 0; 4 5.7 0 0; 5 9.2 0 0; 6 13.5 0 0; 7 0 0 4.5; 8 9.2 0 4.5;
9 13.5 0 4.5; 10 0 0 7.4; 11 9.2 0 7.4; 12 13.5 0 7.4; 14 0 0 11.9;
17 5.7 0 11.9; 18 9.2 0 11.9; 19 13.5 0 11.9; 22 -2 0 6.025; 23 0 -1.2 0;
26 5.7 -1.2 0; 27 9.2 -1.2 0; 28 13.5 -1.2 0; 29 0 -1.2 4.5; 30 9.2 -1.2 4.5;
31 13.5 -1.2 4.5; 32 0 -1.2 7.4; 33 9.2 -1.2 7.4; 34 13.5 -1.2 7.4;
35 0 -1.2 11.9; 38 5.7 -1.2 11.9; 39 9.2 -1.2 11.9; 40 13.5 -1.2 11.9;
41 -2 -1.2 6.025; 42 0 6.2 0; 45 5.7 6.2 0; 46 9.2 6.2 0; 47 13.5 6.2 0;
48 0 6.2 4.5; 49 9.2 6.2 4.5; 50 13.5 6.2 4.5; 51 0 6.2 7.4; 52 9.2 6.2 7.4;
53 13.5 6.2 7.4; 54 0 6.2 11.9; 57 5.7 6.2 11.9; 58 9.2 6.2 11.9;
59 13.5 6.2 11.9; 110 -2 2 6.025; 111 0 2 7.4; 112 0 2 4.5; 113 0 0 5.075;
114 0 2 5.075; 115 0 0 6.975; 116 0 2 6.975; 117 6.2 6.2 4.5; 118 6.2 6.2 7.4;
119 3 6.2 4.5; 120 3 6.2 7.4;
MEMBER INCIDENCES
1 1 4; 4 4 5; 5 5 6; 6 1 7; 7 5 8; 8 6 9; 11 7 113; 12 8 11; 13 9 12; 17 14 17;
20 17 18; 21 18 19; 22 14 10; 23 18 11; 24 19 12; 25 1 23; 28 4 26; 29 5 27;
30 6 28; 31 7 29; 32 8 30; 33 9 31; 34 10 32; 35 11 33; 36 12 34; 37 14 35;
40 17 38; 41 18 39; 42 19 40; 43 22 41; 46 1 42; 49 4 45; 50 5 46; 51 6 47;
52 7 112; 53 8 49; 54 9 50; 55 10 111; 56 11 52; 57 12 53; 58 14 54; 61 17 57;
62 18 58; 63 19 59; 64 22 110; 65 42 45; 68 45 46; 69 46 47; 70 42 48;
71 46 49; 72 47 50; 73 48 119; 75 48 51; 76 49 52; 77 50 53; 78 51 120;
80 54 57; 83 57 58; 84 58 59; 85 54 51; 86 58 52; 87 59 53; 197 112 114;
199 113 115; 200 114 116; 201 113 114; 202 115 10; 203 116 111; 204 115 116;
205 114 110; 206 116 110; 207 113 22; 208 115 22; 209 112 48; 210 111 51;
211 117 49; 212 118 52; 213 117 118; 214 119 117; 215 120 118; 216 119 120;
START GROUP DEFINITION
MEMBER
_LB1 65 68 69 80 83 84
_B2 71 76 86 213 216
_GRADEBEAMS 1 4 TO 8 11 TO 13 17 20 TO 24 199 202
_B3 70 72 75 77 85 87
_LB2 73 78 211 212 214 215
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
END DEFINE MATERIAL
```

```
MEMBER PROPERTY AMERICAN
7 12 23 25 28 29 33 36 37 40 TO 42 46 49 TO 51 54 57 58 61 TO 63 65 -
68 TO 70 72 75 77 80 83 TO 85 87 PRIS YD 0.5 ZD 0.25
31 32 34 35 52 53 55 56 209 210 PRIS YD 0.55 ZD 0.3
MEMBER PROPERTY EUROPEAN
200 201 204 TO 208 PRIS YD 0.4 ZD 0.2
MEMBER PROPERTY EUROPEAN
43 64 197 203 PRIS YD 0.2 ZD 0.2
MEMBER PROPERTY EUROPEAN
71 76 86 213 216 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
1 4 TO 6 8 11 13 17 20 TO 22 24 30 199 202 PRIS YD 0.6 ZD 0.25
MEMBER PROPERTY AMERICAN
73 78 211 212 214 215 PRIS YD 0.75 ZD 0.3
CONSTANTS
BETA 90 MEMB 25 30 TO 37 42 46 51 TO 58 63 201 204 209 210
MATERIAL CONCRETE ALL
*********************************************************************************
SUPPORTS
23 26 TO 35 38 TO 41 FIXED
MEMBER CRACKED
25 28 TO 37 40 TO 43 46 49 TO 58 61 TO 64 201 204 209 -
210 REDUCTION RIY 0.700000 RIZ 0.700000
1 4 TO 8 11 TO 13 17 20 TO 24 65 68 TO 72 75 TO 77 80 83 TO 87 197 199 200 -
202 203 REDUCTION RIX 0.350000 RIY 0.350000 RIZ 0.350000
MEMBER OFFSET
68 69 83 84 START 0.25 0 0
65 68 83 END -0.25 0 0
65 73 78 80 START 0.125 0 0
69 80 84 211 212 END -0.125 0 0
85 87 END 0 0 0.375
70 72 76 START 0 0 0.25
86 87 START 0 0 -0.25
70 72 75 TO 77 END 0 0 -0.25
75 77 START 0 0 0.375
86 END 0 0 0.25
71 END 0 0 0.125
DEFINE IBC 2003
SDS 0.106 SD1 0.032 S1 0.02 IE 1 RX 2.5 RZ 2.5 SCLASS 4
SELFWEIGHT 1
MEMBER WEIGHT
65 68 80 83 UNI 13
73 78 211 212 214 215 UNI 21.8
1 4 TO 6 8 11 13 17 20 TO 22 24 199 202 UNI 23.2
7 12 23 UNI 14
71 72 76 77 86 87 UNI 12.36
213 TO 216 UNI 5
DEFINE WIND LOAD
TYPE 1 WIND X WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 15.000 M 2.000 0.010 0 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.883 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.67646 1.67646 1.68156 1.68654 1.69139 1.69613 1.70075 1.70527 1.70969 -
 1.71401 1.71824 1.72239 1.72645 1.73043 1.73434 HEIG  0  4.572  4.72031 -
 4.86862  5.01692  5.16523  5.31354  5.46185  5.61015  5.75846  5.90677 -
 6.05508  6.20338  6.35169  6.5
```

```
TYPE 2 WIND X LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 15.000 M 2.000 0.010 1 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.883 -0.433 0.550
!> END GENERATED DATA BLOCK
INT -1.28738 -1.28738 HEIG  0  6.5
TYPE 3 WIND X SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 15.000 M 2.000 0.010 2 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.58044 -1.58044 HEIG  0  6.5
TYPE 4 WIND Z WINDWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 15.000 M 20.000 M 2.000 0.010 0 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 0.800 -0.550
!> END GENERATED DATA BLOCK
INT 1.64206 1.64206 1.64697 1.65176 1.65643 1.66099 1.66544 1.66979 1.67404 -
 1.6782 1.68228 1.68627 1.69017 1.69401 1.69777 HEIG  0  4.572  4.72031 -
 4.86862  5.01692  5.16523  5.31354  5.46185  5.61015  5.75846  5.90677 -
 6.05508  6.20338  6.35169  6.5
TYPE 5 WIND Z LEEWARD
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 15.000 M 20.000 M 2.000 0.010 1 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
0 0 0 0.878 -0.500 0.550
!> END GENERATED DATA BLOCK
INT -1.36505 -1.36505 HEIG  0  6.5
TYPE 6 WIND Z SIDE WALL
<! STAAD PRO GENERATED DATA DO NOT MODIFY !!!
ASCE-7-2002:PARAMS 177.000 KMPH 0 3 1 0 0.000 FT 0.000 FT 0.000 FT 1 -
1 6.500 M 20.000 M 40.000 M 2.000 0.010 2 -
0 0 1 0 1.095 1.000 1.000 0.850 0 -
1 0 0 0.850 -0.700 0.550
!> END GENERATED DATA BLOCK
INT -1.58044 -1.58044 HEIG  0  6.5
LOAD 1 LOADTYPE Seismic  TITLE EQK LOAD Z
IBC LOAD Z 1
PERFORM ANALYSIS
CHANGE
LOAD 2 LOADTYPE Seismic  TITLE EQK LOAD X
IBC LOAD X 1
PERFORM ANALYSIS
CHANGE
PERFORM ANALYSIS
CHANGE
LOAD 3 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
65 68 80 83 UNI GY -13
73 78 211 212 214 215 UNI GY -21.8
1 4 TO 6 8 11 13 17 20 TO 22 24 199 202 UNI GY -23.2
```

70 75 85 UMOM GZ 1
72 77 87 UMOM GZ -1
80 83 84 UMOM GX 1
65 68 69 UMOM GX -1
7 12 23 UNI GY -14
71 72 76 77 86 87 UNI GY -12.36
213 TO 216 UNI GY -5
LOAD 4 LOADTYPE Live TITLE LL
MEMBER LOAD
65 68 69 71 72 76 77 80 83 84 86 87 UNI GY -1.125
73 78 211 212 214 215 UNI GY -2
65 68 69 UMOM GX -0.18
80 83 84 UMOM GX 0.18
70 75 85 UMOM GZ 0.18
72 77 87 UMOM GZ -0.18
LOAD 5 LOADTYPE Wind TITLE WL+X
WIND LOAD X 1 TYPE 1 YR 0 7
WIND LOAD -X 1 TYPE 2 YR 0 7
WIND LOAD -Z 1 TYPE 3 YR 0 7
WIND LOAD -Z -1 TYPE 3 YR 0 7
LOAD 6 LOADTYPE Wind TITLE WL+Z
WIND LOAD Z 1 TYPE 4 YR 0 7
WIND LOAD -Z 1 TYPE 5 YR 0 7
WIND LOAD -X 1 TYPE 6 YR 0 7
WIND LOAD -X -1 TYPE 6 YR 0 7
LOAD 7 LOADTYPE Wind TITLE WL-X
WIND LOAD X -1 TYPE 1 YR 0 7
WIND LOAD -X -1 TYPE 2 YR 0 7
WIND LOAD -Z 1 TYPE 3 YR 0 7
WIND LOAD -Z -1 TYPE 3 YR 0 7
LOAD 8 LOADTYPE Wind TITLE WL-Z
WIND LOAD Z -1 TYPE 4 YR 0 7
WIND LOAD -Z -1 TYPE 5 YR 0 7
WIND LOAD -X 1 TYPE 6 YR 0 7
WIND LOAD -X -1 TYPE 6 YR 0 7
LOAD 9 LOADTYPE Temperature TITLE TEMP
TEMPERATURE LOAD
46 49 TO 58 61 TO 65 68 TO 73 75 TO 78 80 83 TO 87 197 200 201 203 TO 206 -
209 TO 216 TEMP 25 25 25
LOAD COMB 10 SLS: (DL)
3 1.0
LOAD COMB 11 SLS: (DL+T)
3 1.0
LOAD COMB 12 SLS: (D+LR)
3 1.0 4 1.0
LOAD COMB 13 SLS: (D+0.75LR)
3 1.0 4 0.75
LOAD COMB 14 SLS: (D+0.7EZ)
3 1.0 1 0.7
LOAD COMB 15 SLS: (D+0.7EX)
3 1.0 2 0.7
LOAD COMB 16 SLS: (D+0.75WX1+0.75LR)
3 1.0 5 0.75 4 0.75
LOAD COMB 17 SLS: (D+0.75WZ1+0.75LR)
3 1.0 6 0.75 4 0.75
LOAD COMB 18 SLS: (0.6D+WX1)
3 0.6 5 1.0
LOAD COMB 19 SLS: (0.6D+WZ1)
3 0.6 6 1.0

LOAD COMB 20 SLS: (0.6D+0.7EX)
3 0.6 2 0.7
LOAD COMB 21 SLS: (0.6D+0.7EZ)
3 0.6 1 0.7
LOAD COMB 22 SLS: (D+WX1)
3 1.0 5 1.0
LOAD COMB 23 SLS: (D+WZ1)
3 1.0 6 1.0
LOAD COMB 24 ULS: (1.4 DL)
3 1.4
LOAD COMB 25 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 26 ULS: (1.4D+1.4T+0.5LR)
3 1.4 4 0.5
LOAD COMB 27 ULS: (1.2D+0.80WX1+1.6LR)
3 1.2 5 0.8 4 1.6
LOAD COMB 28 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 6 1.6 4 0.5
LOAD COMB 29 ULS: (1.2D+1.6WX1+0.5LR)
3 1.2 4 0.5 5 1.6
LOAD COMB 30 ULS: (1.2D+1.6WZ1+0.5LR)
3 1.2 4 0.5 6 1.6
LOAD COMB 31 ULS: (1.2D+EX)
3 1.2 2 1.0
LOAD COMB 32 ULS: (1.2D+EZ)
3 1.2 1 1.0
LOAD COMB 33 ULS: (0.90D+1.6WX1)
3 0.9 5 1.6
LOAD COMB 34 ULS: (0.90D+1.6WZ1)
3 0.9 6 1.6
LOAD COMB 35 ULS: (0.90D+EZ)
3 0.9 1 1.0
LOAD COMB 36 ULS: (0.90D+EX)
3 0.9 2 1.0
LOAD COMB 39 SLS: (D+0.75WX2+0.75LR)
3 1.0 7 0.75 4 0.75
LOAD COMB 40 SLS: (D+0.75WZ2+0.75LR)
3 1.0 8 0.75 4 0.75
LOAD COMB 41 SLS: (0.6D+WX2)
3 0.6 7 1.0
LOAD COMB 42 SLS: (0.6D+WZ2)
3 0.6 8 1.0
LOAD COMB 45 SLS: (D+WX2)
3 1.0 7 1.0
LOAD COMB 46 SLS: (D+WZ2)
3 1.0 8 1.0
LOAD COMB 47 ULS: (1.2D+0.80WX2+1.6LR)
3 1.2 7 0.8 4 1.6
LOAD COMB 48 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 8 1.6 4 0.5
LOAD COMB 49 ULS: (1.2D+1.6WX2+0.5LR)
3 1.2 4 0.5 7 1.6
LOAD COMB 50 ULS: (1.2D+1.6WZ2+0.5LR)
3 1.2 4 0.5 8 1.6
LOAD COMB 53 ULS: (0.90D+1.6WX2)
3 0.9 7 1.6
LOAD COMB 54 ULS: (0.90D+1.6WZ2)
3 0.9 8 1.6
PERFORM ANALYSIS

```
START CONCRETE DESIGN
CODE ACI
CLB 0.04 MEMB 1 4 TO 8 11 TO 13 17 20 TO 25 28 TO 37 40 TO 43 46 49 TO 58 -
61 TO 65 68 TO 72 75 TO 77 80 83 TO 87 197 199 TO 210
CLS 0.04 MEMB 1 4 TO 8 11 TO 13 17 20 TO 25 28 TO 37 40 TO 43 46 49 TO 58 -
61 TO 65 68 TO 72 75 TO 77 80 83 TO 87 197 199 TO 210 213 216
CLT 0.04 MEMB 1 4 TO 8 11 TO 13 17 20 TO 25 28 TO 37 40 TO 43 46 49 TO 58 -
61 TO 65 68 TO 72 75 TO 77 80 83 TO 87 197 199 TO 210
MINSEC 10 MEMB 1 4 TO 8 11 TO 13 17 20 TO 25 28 TO 37 40 TO 43 46 49 TO 58 -
61 TO 65 68 TO 72 75 TO 77 80 83 TO 87 197 199 TO 210 213 216
MINMAIN 20 MEMB 71 73 76 78 86 211 212 214 215
MAXMAIN 20 MEMB 71 76 86 213 216
MINMAIN 16 MEMB 1 4 TO 8 11 TO 13 17 20 TO 24 65 68 TO 70 72 75 77 80 -
83 TO 85 87 199 202
MAXMAIN 16 MEMB 1 4 TO 8 11 TO 13 17 20 TO 24 65 68 TO 70 72 75 77 80 -
83 TO 85 87 199 202
SFACE 0.5 MEMB 65 68 TO 70 72 73 75 77 78 80 83 TO 85 87 211 212 214 215
SFACE 0.5 MEMB 71 76 86 213 216
EFACE 0.5 MEMB 65 68 69 73 78 80 83 84 211 212 214 215
EFACE 0.5 MEMB 71 76 86 213 216
DESIGN BEAM 1 4 TO 8 11 TO 13 17 20 TO 24 65 68 TO 73 75 TO 78 80 83 TO 87 -
199 202 211 TO 216
DESIGN COLUMN 25 28 TO 37 40 TO 43 46 49 TO 58 61 TO 64 201 204 209 210
END CONCRETE DESIGN
PARAMETER 2
CODE AISC UNIFIED 2005
FINISH
```

ATTACHMENT 2 FOUNDATION DESIGN CALCULATIONS

PROPRIETARY & CONFIDENTIAL

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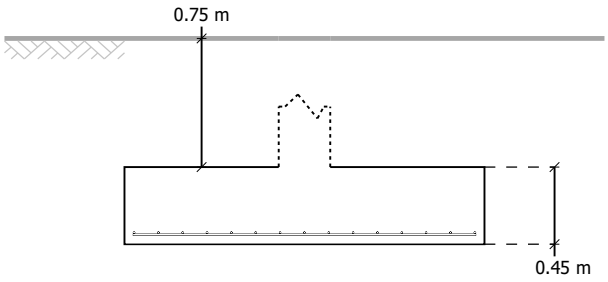
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Isolated Footing Design(ACI 318-08)

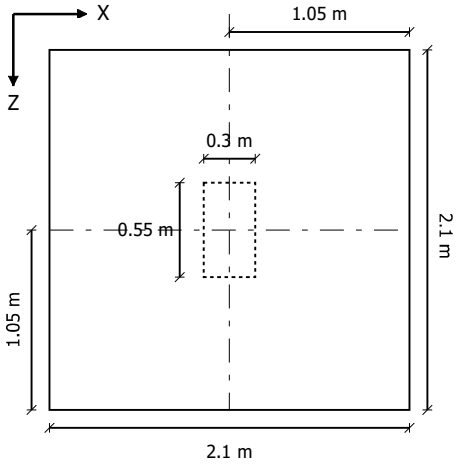
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
30	1	2.100m	2.100m	0.450m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
30	15 - ϕ12	15 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F1



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 450.000mm
Footing Length - X (Fl) : 2100.000mm
Footing Width - Z (Fw) : 2100.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.300m

Column Width - Z (B_{col}) : 0.550m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 750.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 2.100m

Initial Width (W_o) = 2.100m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	290.512	8.280	13.479	4.521	-21.078
11	290.512	8.280	13.479	4.521	-21.078
12	303.458	8.779	13.207	4.452	-22.340
13	300.221	8.654	13.275	4.469	-22.024
14	276.435	8.381	20.867	22.174	-21.377
15	293.796	12.214	13.520	4.613	-34.404
16	302.938	12.676	11.943	3.571	-37.847
17	267.716	6.713	38.857	50.820	-13.658
18	177.930	10.330	6.312	1.515	-33.743

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
19	130.967	2.379	42.196	64.514	-1.492
20	177.592	8.902	8.128	2.805	-25.972
21	160.230	5.069	15.475	20.366	-12.946
22	294.134	13.642	11.703	3.323	-42.174
23	247.172	5.691	47.588	66.322	-9.923
39	296.191	4.766	11.894	3.488	-7.924
40	329.236	5.210	-12.171	-41.976	-11.324
41	168.933	-0.216	6.246	1.404	6.154
42	212.993	0.376	-25.840	-59.215	1.621
45	285.138	3.096	11.638	3.212	-2.277
46	329.198	3.688	-20.448	-57.406	-6.810

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	290.512	8.280	13.479	4.521	-21.078
24	406.717	11.592	18.871	6.330	-29.509
25	413.190	11.841	18.735	6.295	-30.140
26	413.190	11.841	18.735	6.295	-30.140
27	372.226	15.024	14.319	4.357	-44.190
28	285.743	6.043	70.613	104.272	-8.076
29	360.883	18.765	13.198	3.474	-59.679
30	285.743	6.043	70.613	104.272	-8.076
31	353.306	15.556	16.233	5.557	-44.330
32	328.505	10.080	26.728	30.644	-25.722
33	267.257	16.031	9.290	2.152	-52.725
34	192.116	3.310	66.705	102.950	-1.122
35	241.351	7.596	22.685	29.288	-19.398
36	266.153	13.072	12.189	4.201	-38.007
47	365.029	6.587	14.266	4.268	-12.273
48	416.985	2.838	-38.245	-93.693	-3.096
49	346.489	1.891	13.092	3.296	4.156
50	416.985	2.838	-38.245	-93.693	-3.096
53	252.863	-0.843	9.185	1.974	11.111
54	323.358	0.105	-42.153	-95.015	3.858

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 4.410 m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.746 m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 2.100 m Governing Load Case : # 10

Width (W_2) = 2.100 m Governing Load Case : # 10

Depth (D_2) = 0.450 m Governing Load Case : # 48

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 4.410 m^2

Final Soil Height = 0.750 m

Footing Self Weight = 47.626 kN

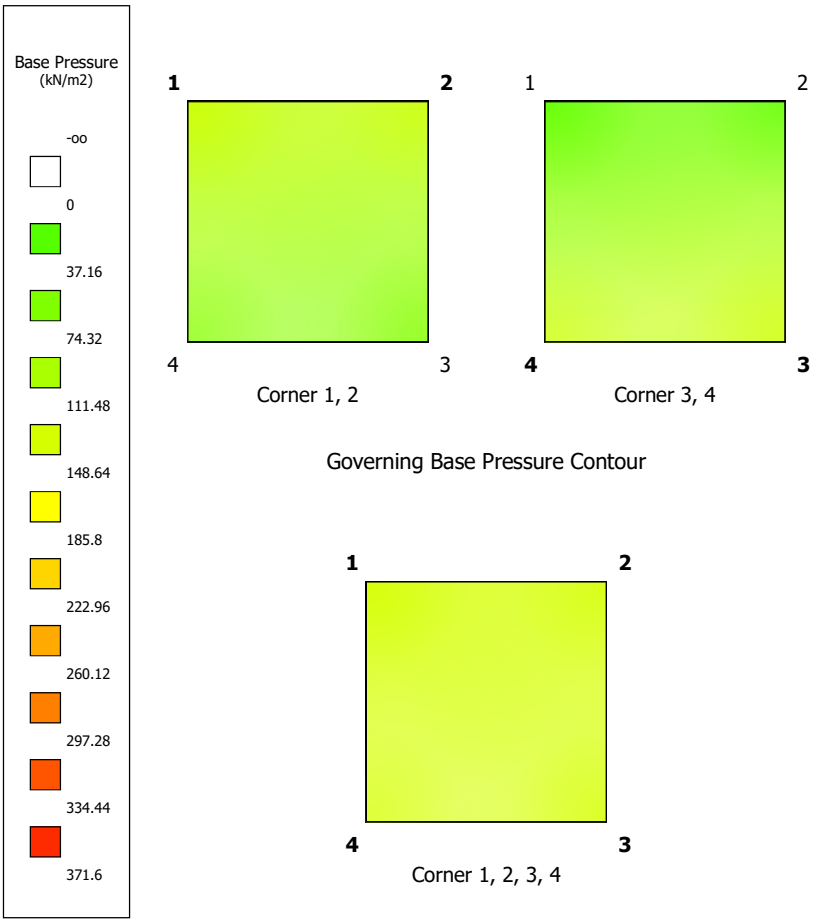
Gross Soil Bearing Capacity = 371.60 kN/m2

Soil Weight On Top Of Footing = 57.305 kN

Pressures at 4 Corners

Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner



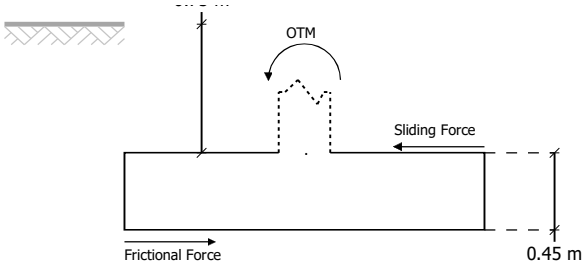
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
46	**136.1092**	147.0840	60.7750	49.8001	0.000
46	136.1092	**147.0840**	60.7750	49.8001	0.000
23	14.9105	31.0865	**144.7736**	128.5976	0.000
23	14.9105	31.0865	144.7736	**128.5976**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
46	**136.1092**	147.0840	60.7750	49.8001
46	136.1092	**147.0840**	60.7750	49.8001
23	14.9105	31.0865	**144.7736**	128.5976
23	14.9105	31.0865	144.7736	**128.5976**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	25.313	15.549	18.737	1.500	39.237	16.808	1.500
11	25.313	15.549	18.737	1.500	39.237	16.808	1.500
12	24.655	16.389	19.302	1.500	41.268	16.375	1.500
13	24.812	16.175	19.163	1.500	40.754	16.479	1.500
14	24.118	9.687	12.712	1.500	12.692	15.990	1.500
15	17.302	15.631	16.403	1.500	39.155	10.535	1.500
16	17.054	18.100	17.554	1.500	47.897	9.872	1.500
17	29.423	5.083	7.083	1.500	5.731	23.561	1.500
18	14.513	23.752	17.514	1.500	68.241	7.780	1.500
19	52.551	2.963	4.184	1.500	2.969	97.334	1.500
20	16.821	18.422	17.567	1.500	45.932	9.952	1.500
21	27.726	9.082	12.205	1.500	10.194	18.396	1.500
22	15.504	18.072	16.641	1.500	48.803	8.708	1.500
23	32.791	3.921	5.507	1.500	4.216	29.751	1.500
39	44.606	17.875	23.465	1.500	47.668	41.998	1.500
40	44.164	18.907	24.581	1.500	9.611	33.477	1.500
41	670.866	23.239	32.846	1.500	68.282	46.273	1.500
42	448.050	6.521	9.221	1.500	4.715	231.115	1.500
45	66.785	17.765	24.279	1.500	48.499	112.050	1.500
46	62.389	11.252	15.660	1.500	6.846	54.020	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18
Governing Disturbing Force : 10.330kN
Governing Restoring Force : 149.917kN
Minimum Sliding Ratio for the Critical Load Case : 14.513
Critical Load Case for Overturning about X-Direction : 19
Governing Overturning Moment : 83.502kNm
Governing Resisting Moment : 247.884kNm
Minimum Overturning Ratio for the Critical Load Case : 2.969

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19
Governing Disturbing Force : 42.196kN
Governing Restoring Force : 125.026kN

Minimum Sliding Ratio for the Critical Load Case : 2.963

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -38.392kNm

Governing Resisting Moment : 298.700kNm

Minimum Overturning Ratio for the Critical Load Case : 7.780

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 19

Governing Disturbing Force : 42.263kN

Governing Restoring Force : 176.814kN

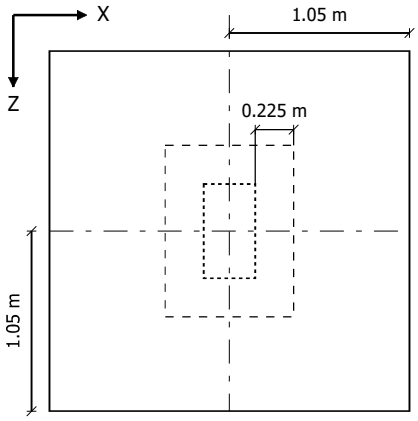
Minimum Sliding Ratio for the Critical Load Case : 4.184

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.450m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.394m

For rectangular column, β_c = B_{col} / D_{col} = 1.833

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 444.382kN, Load Case # 48

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 3.276m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2651.544kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 4318.454kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 2536.260kN

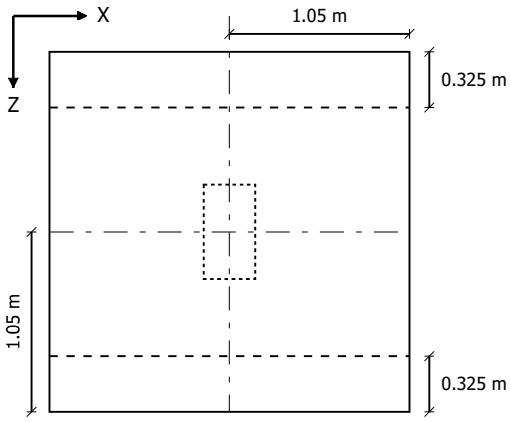
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1902.195kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 812.904kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.381m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

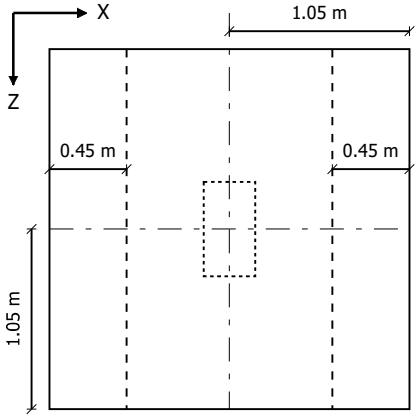
From above calculations, 0.75 X V_c = 609.678 kN

Critical load case for V_{ux} is # 48 $V_{ux} = V_{ux}|_{x=D_x}$ = 141.750 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 812.904 kN

Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.506 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

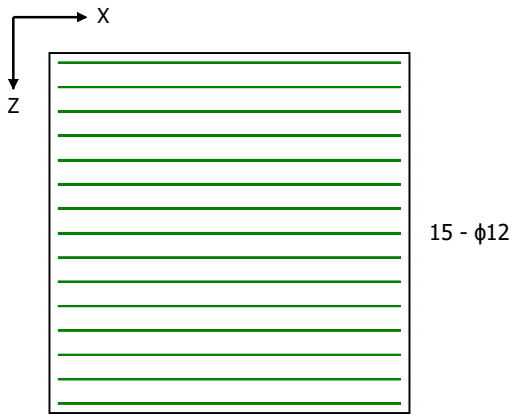
From above calculations, 0.75 X V_c = 609.678 kN

Critical load case for V_{uz} is # 29 $V_{uz} = V_{uz}|_{z=D_z}$ = 147.838 kN

0.75 X V_c > V_{uz} hence, OK

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.394 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00167
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.200 m
Ultimate moment,	$M_u\big\vert_{z=D_z}$ =	116.667 kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	129.630 kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00095
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00083
Since	ρ= ρ_{min}	ρ_{min} Governs
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1579.712 mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = =0.410 m

$\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ =

Available development length for bars, D_L = $0.5 \times (L - D_{col}) \cdot - C_{cover}$ = 0.850 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.394 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00205

From ACI Cl.7.6.1, minimum req'd clear distance between bars

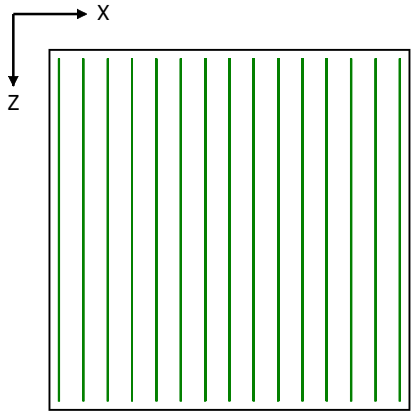
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.074

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



15 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 48

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.382 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = $0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$ 0.03317

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00164

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 0.775 m

Ultimate moment, $M_u|_{x=D_x}$ = 108.792 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 120.880 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ = 0.00095

(Based on gross depth) ρ x d_{eff} / Depth = 0.00080

Since $\rho = \rho_{min}$ ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1547.059 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 15

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1696.448 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.382 m

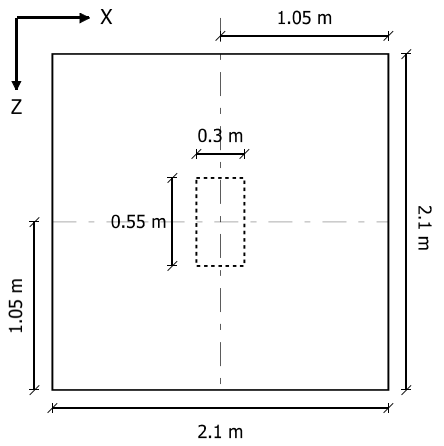
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00211

From ACI Cl.7.6.1, minimum req'd clear distance between bars

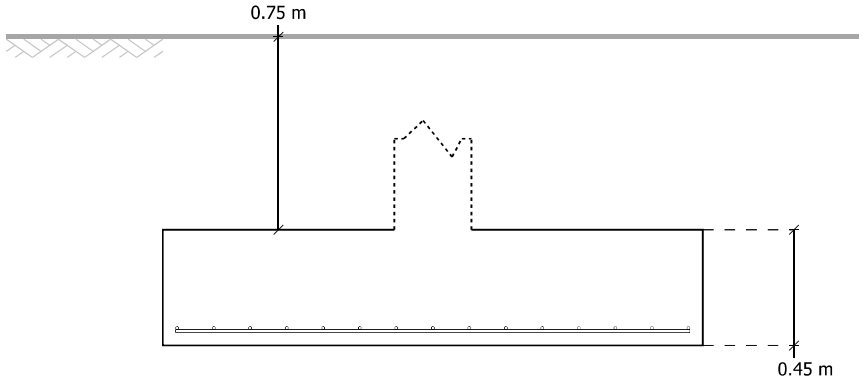
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.097

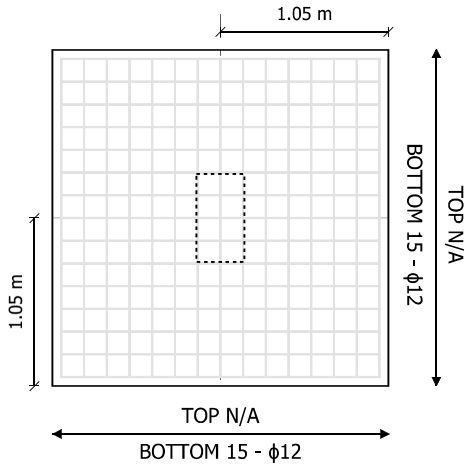
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	2.1 m	2.1 m	0.45 m	N/A	N/A	15 - ϕ12	15 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

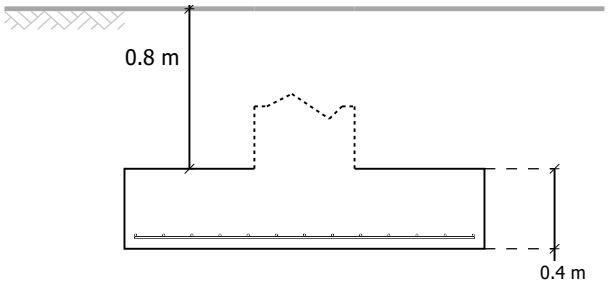
Print Calculation Sheet

Isolated Footing Design(ACI 318-08)

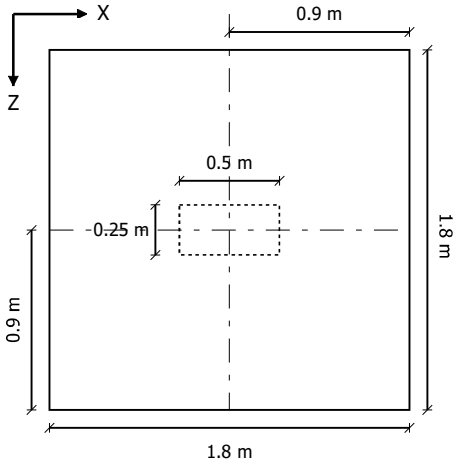
Footing No.	Group ID	Foundation Geometry		
-	-	Length	Width	Thickness
26	1	1.800m	1.800m	0.400m

Footing No.	Footing Reinforcement				Pedestal Reinforcement	
-	Bottom Reinforcement(M_z)	Bottom Reinforcement(M_x)	Top Reinforcement(M_z)	Top Reinforcement(M_x)	Main Steel	Trans Steel
26	13 - ϕ12	13 - ϕ12	N/A	N/A	N/A	N/A

FOUNDATION-F2



Elevation



Plan

Input Values

Footing Geomtery

Design Type : Set Dimension
Footing Thickness (Ft) : 400.000mm
Footing Length - X (Fl) : 1800.000mm
Footing Width - Z (Fw) : 1800.000mm
Eccentricity along X (Oxd) : 0.000mm
Eccentricity along Z (Ozd) : 0.000mm

Column Dimensions

Column Shape : Rectangular
Column Length - X (D_{col}) : 0.500m

Column Width - Z (B_{col}) : 0.250m

Pedestal

Include Pedestal? No
Pedestal Shape : N/A
Pedestal Height (Ph) : N/A
Pedestal Length - X (Pl) : N/A
Pedestal Width - Z (Pw) : N/A

Design Parameters

Concrete and Rebar Properties

Unit Weight of Concrete : 24.000kN/m3
Strength of Concrete : 35.000N/mm2
Yield Strength of Steel : 420.000N/mm2
Minimum Bar Size : #12
Maximum Bar Size : #12
Top Footing Minimum Bar Size : #12
Top Footing Maximum Bar Size : #12
Pedestal Minimum Bar Size : #16
Pedestal Maximum Bar Size : #16
Minimum Bar Spacing : 125.000mm
Maximum Bar Spacing : 150.000mm
Pedestal Clear Cover (P, CL) : 50.000mm
Bottom Footing Clear Cover (F, CL) : 50.000mm

Soil Properties

Soil Type : Cohesionless Soil
Unit Weight : 18.000kN/m3
Soil Bearing Capacity : 350.000kN/m2
Soil Bearing Capacity Type: Net Bearing Capacity
Soil Surcharge : 0.000kN/m2
Depth of Soil above Footing : 800.000mm
Type of Depth : Fixed Top
Undrained Shear Strength : 0.000kN/m2
Bearing Capacity Input Method: Fixed Bearing Capacity

Sliding and Overturning

Coefficient of Friction : 0.530
Factor of Safety Against Sliding : 1.500
Factor of Safety Against Overturning : 1.500

Global Settings

Top Reinforcement Option : Calculate only when foundation is subjected to uplift forces
Concrete Design Option : Gross Pressure
Top Reinforcement Factor : 1.000

Design Calculations

Footing Size

Initial Length (L_o) = 1.800m

Initial Width (W_o) = 1.800m

Load Combinations

Load Combination/s- Service Stress Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
11	SLS: (DL+T)	1.00	1.00	1.00
12	SLS: (D+LR)	1.00	1.00	1.00
13	SLS: (D+0.75LR)	1.00	1.00	1.00
14	SLS: (D+0.7EZ)	1.00	1.00	1.00
15	SLS: (D+0.7EX)	1.00	1.00	1.00
16	SLS: (D+0.75WX1+0.75LR)	1.00	1.00	1.00
17	SLS: (D+0.75WZ1+0.75LR)	1.00	1.00	1.00
18	SLS: (0.6D+WX1)	1.00	1.00	1.00
19	SLS: (0.6D+WZ1)	1.00	1.00	1.00
20	SLS: (0.6D+0.7EX)	1.00	1.00	1.00
21	SLS: (0.6D+0.7EZ)	1.00	1.00	1.00
22	SLS: (D+WX1)	1.00	1.00	1.00
23	SLS: (D+WZ1)	1.00	1.00	1.00
39	SLS: (D+0.75WX2+0.75LR)	1.00	1.00	1.00
40	SLS: (D+0.75WZ2+0.75LR)	1.00	1.00	1.00
41	SLS: (0.6D+WX2)	1.00	1.00	1.00
42	SLS: (0.6D+WZ2)	1.00	1.00	1.00
45	SLS: (D+WX2)	1.00	1.00	1.00
46	SLS: (D+WZ2)	1.00	1.00	1.00

Load Combination/s- Strength Level				
Load Combination Number	Load Combination Title	Load Combination Factor	Soil Bearing Factor	Self Weight Factor
10	SLS: (DL)	1.00	1.00	1.00
24	ULS: (1.4 DL)	1.00	1.00	1.00
25	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
26	ULS: (1.4D+1.4T+0.5LR)	1.00	1.00	1.00
27	ULS: (1.2D+0.80WX1+1.6LR)	1.00	1.00	1.00
28	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
29	ULS: (1.2D+1.6WX1+0.5LR)	1.00	1.00	1.00
30	ULS: (1.2D+1.6WZ1+0.5LR)	1.00	1.00	1.00
31	ULS: (1.2D+EX)	1.00	1.00	1.00
32	ULS: (1.2D+EZ)	1.00	1.00	1.00
33	ULS: (0.90D+1.6WX1)	1.00	1.00	1.00
34	ULS: (0.90D+1.6WZ1)	1.00	1.00	1.00
35	ULS: (0.90D+EZ)	1.00	1.00	1.00
36	ULS: (0.90D+EX)	1.00	1.00	1.00
47	ULS: (1.2D+0.80WX2+1.6LR)	1.00	1.00	1.00
48	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
49	ULS: (1.2D+1.6WX2+0.5LR)	1.00	1.00	1.00
50	ULS: (1.2D+1.6WZ2+0.5LR)	1.00	1.00	1.00
53	ULS: (0.90D+1.6WX2)	1.00	1.00	1.00
54	ULS: (0.90D+1.6WZ2)	1.00	1.00	1.00

Applied Loads

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	234.187	34.664	0.205	1.184	-11.898
11	234.187	34.664	0.205	1.184	-11.898
12	239.554	34.221	0.273	1.319	-11.818
13	238.212	34.332	0.256	1.285	-11.838
14	233.931	34.806	1.680	5.198	-12.106
15	226.951	39.978	0.154	1.071	-20.212
16	224.431	47.871	-17.249	-19.569	-30.184
17	234.862	35.612	19.152	27.154	-14.916
18	122.137	38.850	-23.217	-27.095	-31.600

Applied Loads - Service Stress Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
19	136.045	22.505	25.317	35.203	-11.243
20	133.277	26.113	0.072	0.598	-15.453
21	140.256	20.941	1.598	4.725	-7.347
22	215.811	52.716	-23.135	-26.622	-36.360
23	229.719	36.371	25.399	35.677	-16.002
39	253.051	20.324	-16.697	-18.273	7.529
40	236.403	34.800	-16.053	-22.193	-13.611
41	160.297	2.122	-22.480	-25.367	18.684
42	138.100	21.422	-21.623	-30.594	-9.502
45	253.972	15.987	-22.398	-24.893	13.924
46	231.774	35.287	-21.541	-30.120	-14.261

Applied Loads - Strength Level					
LC	Axial (kN)	Shear X (kN)	Shear Z (kN)	Moment X (kNm)	Moment Z (kNm)
10	234.187	34.664	0.205	1.184	-11.898
24	327.862	48.530	0.287	1.657	-16.657
25	330.545	48.308	0.321	1.725	-16.617
26	330.545	48.308	0.321	1.725	-16.617
27	274.912	55.330	-18.318	-20.608	-33.719
28	276.560	44.106	40.591	56.676	-20.805
29	254.307	70.259	-37.065	-43.001	-53.376
30	276.560	44.106	40.591	56.676	-20.805
31	270.688	49.189	0.174	1.260	-26.154
32	280.658	41.800	2.353	7.156	-14.574
33	181.367	60.081	-37.160	-43.423	-49.847
34	203.620	33.929	40.496	56.254	-17.275
35	210.402	31.401	2.292	6.801	-11.005
36	200.432	38.789	0.112	0.904	-22.585
47	305.440	25.947	-17.728	-19.226	6.508
48	279.848	42.373	-34.513	-48.598	-18.019
49	315.364	11.492	-35.886	-40.236	27.078
50	279.848	42.373	-34.513	-48.598	-18.019
53	242.424	1.315	-35.981	-40.658	30.608
54	206.908	32.195	-34.608	-49.021	-14.490

Reduction of force due to buoyancy = 0.000kN

Effect due to adhesion = 0.000kN

Area from initial length and width, $A_o = L_o \times W_o = 3.240m^2$

Min. area required from bearing pressure, $A_{min} = P / q_{max} = 1.622m^2$

Note: A_{min} is an initial estimation.
P = Critical Factored Axial Load(without self weight/buoyancy/soil).
q_{max} = Respective Factored Bearing Capacity.

Final Footing Size

Length (L_2) = 1.800 m Governing Load Case : # 10

Width (W_2) = 1.800 m Governing Load Case : # 10

Depth (D_2) = 0.400 m Governing Load Case : # 29

Depth is governed by Ultimate Load Case

(Service check is performed with footing thickness requirements from concrete check)

Area (A_2) = 3.240 m^2

Final Soil Height = 0.800 m

Footing Self Weight = 31.103 kN

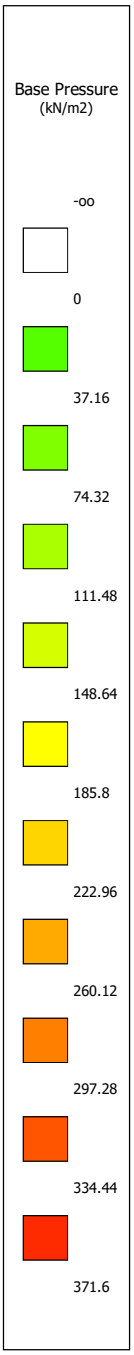
Gross Soil Bearing Capacity = 371.60 kN/m2

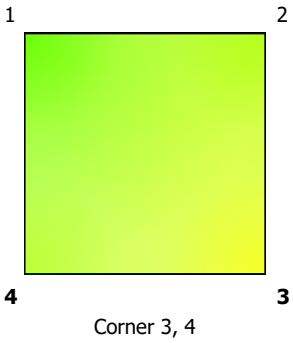
Soil Weight On Top Of Footing = 44.854 kN

Pressures at 4 Corners

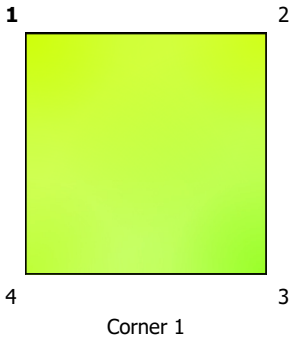
Please note that pressures values displayed in tables below are calculated after dividing by soil bearing factor

Governing Base Pressure Contour for Each Corner

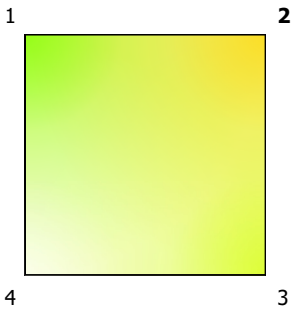




Corner 3, 4

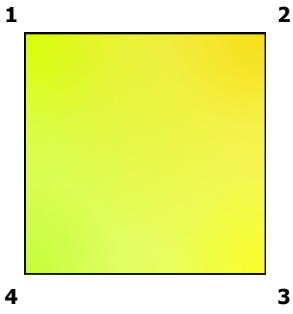


Corner 1



Corner 2

Governing Base Pressure Contour



Corner 1, 2, 3, 4

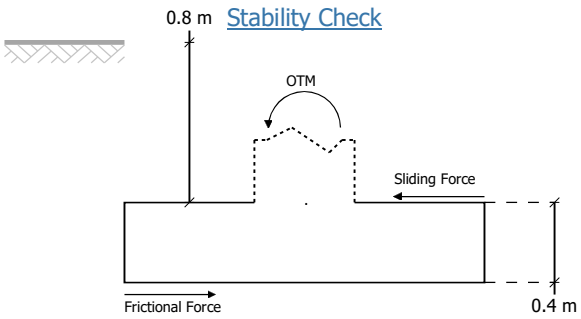
Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)	Area of footing in uplift (A_u) (m^2)
45	**144.4053**	128.9118	59.2548	74.7483	0.000
22	67.8603	**186.0632**	112.2439	-5.9591	0.005
23	15.7564	78.6183	**172.9330**	110.0711	0.000
23	15.7564	78.6183	172.9330	**110.0711**	0.000

If A_u is zero, there is no uplift and no pressure adjustment is necessary. Otherwise, to account for uplift, areas of negative pressure will be set to zero and the pressure will be redistributed to remaining corners.

Summary of Adjusted Pressures at 4 corners Four Corners

Load Case	Pressure at corner 1 (q_1) (kN/m2)	Pressure at corner 2 (q_2) (kN/m2)	Pressure at corner 3 (q_3) (kN/m2)	Pressure at corner 4 (q_4) (kN/m2)
45	**144.4053**	128.9118	59.2548	74.7483
22	67.8472	**186.0738**	112.2329	0.0000
23	15.7564	78.6183	**172.9330**	110.0711
23	15.7564	78.6183	172.9330	**110.0711**

Stability Check



-	Factor of safety against sliding			Factor of safety against overturning			
Load Case No.	Along X-Direction	Along Z-Direction	Resultant	Required FOS	About X-Direction	About Z-Direction	Required FOS
10	4.742	802.126	6.706	1.500	221.459	10.834	1.500
11	4.742	802.126	6.706	1.500	221.459	10.834	1.500
12	4.886	613.253	6.910	1.500	199.737	11.133	1.500
13	4.850	651.088	6.859	1.500	204.693	11.058	1.500
14	4.719	97.759	6.665	1.500	47.715	10.715	1.500
15	4.016	1040.835	5.679	1.500	241.717	7.530	1.500
16	3.326	9.230	4.425	1.500	10.260	5.480	1.500
17	4.626	8.602	5.762	1.500	8.070	9.593	1.500
18	2.702	4.522	3.281	1.500	4.934	3.782	1.500
19	4.993	4.438	4.691	1.500	4.236	9.424	1.500
20	4.247	1534.376	6.006	1.500	302.513	7.271	1.500
21	5.472	71.706	7.717	1.500	36.503	12.376	1.500
22	2.933	6.684	3.799	1.500	7.353	4.571	1.500
23	4.454	6.378	5.165	1.500	6.028	9.005	1.500
39	8.580	10.444	9.375	1.500	11.916	492.691	1.500
40	4.757	10.313	6.109	1.500	9.867	10.211	1.500
41	59.021	5.570	7.842	1.500	6.224	11.922	1.500
42	5.296	5.247	5.271	1.500	4.940	10.661	1.500
45	10.938	7.807	8.986	1.500	8.807	39.435	1.500
46	4.622	7.572	5.579	1.500	7.181	9.760	1.500

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - X Direction

Critical Load Case for Sliding along X-Direction : 18

Governing Disturbing Force : 38.850kN

Governing Restoring Force : 104.990kN

Minimum Sliding Ratio for the Critical Load Case : 2.702

Critical Load Case for Overturning about X-Direction : 19

Governing Overturning Moment : 45.330kNm

Governing Resisting Moment : 192.013kNm

Minimum Overturning Ratio for the Critical Load Case : 4.236

Critical Load Case And The Governing Factor Of Safety For Overturning And Sliding - Z Direction

Critical Load Case for Sliding along Z-Direction : 19

Governing Disturbing Force : 25.317kN

Governing Restoring Force : 112.361kN

Minimum Sliding Ratio for the Critical Load Case : 4.438

Critical Load Case for Overturning about Z-Direction : 18

Governing Overturning Moment : -47.140kNm

Governing Resisting Moment : 178.281kNm

Minimum Overturning Ratio for the Critical Load Case : 3.782

Critical Load Case And The Governing Factor Of Safety For Sliding Along Resultant Direction

Critical Load Case for Sliding along Resultant Direction : 18

Governing Disturbing Force : 45.259kN

Governing Restoring Force : 148.478kN

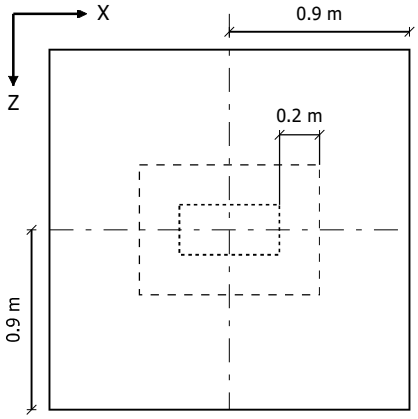
Minimum Sliding Ratio for the Critical Load Case : 3.281

Compression Development Length Check

Development length skipped as column reinforcement is not specified in input (Column Dimnesion Task Pane)

Shear Calculation

Punching Shear Check



Plan

Total Footing Depth, D = 0.400m

Calculated Effective Depth, d_{eff} = D - C_{cover} - 0.5 * d_b = 0.344m

For rectangular column, β_c = B_{col} / D_{col} = 2.000

Effective depth, d_{eff}, increased until $0.75XV_c \geq$ Punching Shear Force

Punching Shear Force, Vu = 343.603kN, Load Case # 25

From ACI Cl.11.11.2, b_o for column= $2 \times (B_{col} + D_{col} + 2 \times d_{eff})$ = 2.876m

Equation 11-31, V_{c1} = $0.17 \times \left(1 + \frac{2}{\beta_c}\right) \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

Equation 11-32, V_{c2} = $0.083 \times \left(\frac{\alpha_s \times d}{b_o} + 2\right) \times \lambda \times \left(\sqrt{f_c'} \times b_o \times d_{eff}\right)$ 3297.265kN

Equation 11-33, V_{c3} = $0.33 \times \lambda \times \sqrt{f_c'} \times b_o \times d_{eff}$ 1944.021kN

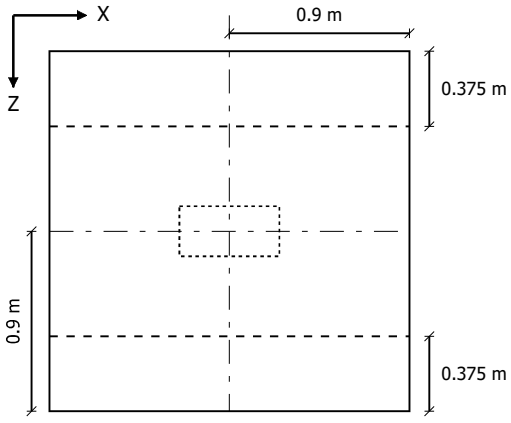
Punching shear strength, V_c = 0.75 X minimum of (V_{c1}, V_{c2}, V_{c3}) = 1458.016kN

0.75 X V_c > V_u hence, OK

One-Way Shear Check

One-Way Shear Along X

(Shear Plane Parallel to Global X Axis)



Plan

From ACI Cl.11.2.1.1, V_c = $2 \times L \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351kN

Distance along X to design for shear, D_x = $0.5 \times (W \pm D_{col}) - d_{eff} + O_{xd}$ = 0.431m

Check that 0.75 X V_c > V_{ux} where V_{ux} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the X axis.

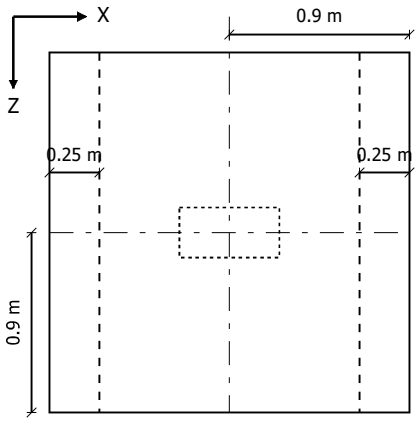
From above calculations, 0.75 X V_c = 456.264 kN

Critical load case for V_{ux} is # 28 $V_{ux} = V_{ux}|_{x=D_x}$ = 128.662 kN

0.75 X V_c > V_{ux} hence, OK

One-Way Shear Along Z

(Shear Plane Parallel to Global Z Axis)



Plan

From ACI Cl.11.2.1.1, V_c =

$2 \times W \times d_{eff} \times \sqrt{1000 \times F_c'}$ = 608.351 kN

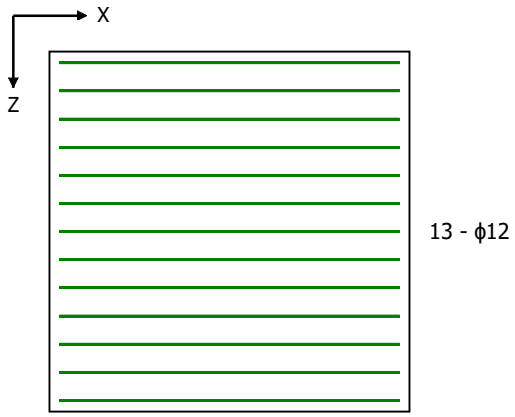
Distance along X to design for shear, D_z = $0.5 \times (L \pm B_{col}) - d_{eff} + O_{xd}$ = 0.306 m

Check that **0.75 X V_c > V_{uz}** where V_{uz} is the shear force for the critical load cases at a distance d_{eff} from the face of the column caused by bending about the Z axis.

From above calculations,	0.75 X V_c =	456.264	kN
Critical load case for V_{uz} is # 29	$V_{uz} = V_{uz}\|_{z=D_z}$ =	92.563	kN
	0.75 X V_c > V_{uz} hence, OK		

Flexure About Z-Axis

Design For Bottom Reinforcement Parallel to X Axis



Calculate the flexural reinforcement along the X direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 29

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} =		0.344 m
Factor β_1 from ACI Cl.10.2.7.3 =		0.796
From ACI Cl. 10.3.2, ρ_{bal} =	$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$	0.03317
From ACI Cl. 10.3.3, ρ_{max} =	$0.75 \times \rho_{bal}$ =	0.02488
From ACI Cl. 7.12.2, ρ_{min} =		0.00163
From Ref. 1, Eq. 3.8.4a, constant m =	$\frac{F_y}{\left(0.85 \times F_c'\right)}$ =	14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about Z axis is performed at the face of the column at a distance, D_x =	$0.5 \times L \pm 0.5 \times D_{col} + O_{xd}$ =	1.150	m
Ultimate moment,	$M_u\|_{z=D_z}$ =	64.975	kNm
Nominal moment capacity, M_n =	$\frac{M_u}{\Phi}$ =	72.194	kNm
(Based on effective depth) Required ρ =	$\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{\left(F_y \times W \times d_{eff}^2\right)}}\right]$ =	0.00081	
	(Based on gross depth) ρ x d_{eff} / Depth =	0.00070	
Since	ρ= ρ_{min}	ρ_{min} Governs	
Area of Steel Required, A_s =	$\rho \times W \times d_{eff}$ =	1175.697	mm2

Selected bar Size = #12

Minimum spacing allowed (S_{min}) = = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f'_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.600 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 0.5 X (dia. of one bar) = 0.344 m

Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00237

From ACI Cl.7.6.1, minimum req'd clear distance between bars

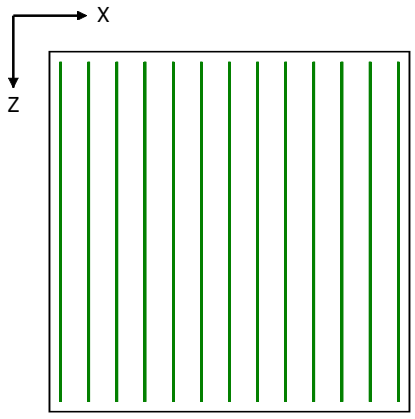
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.251

Check to see if width is sufficient to accomodate bars

Flexure About X-Axis

Design For Bottom Reinforcement Parallel to Z Axis



13 - ϕ12

Calculate the flexural reinforcement along the Z direction of the footing. Find the area of steel required, A, as per Section 3.8 of Reinforced Concrete Design (5th ed.) by Salmon and Wang (Ref. 1)

Critical Load Case # 28

The strength values of steel and concrete used in the formulae are in ksi

Bars parallel to X Direction are placed at bottom

Effective Depth d_{eff} = 0.332 m

Factor β_1 from ACI Cl.10.2.7.3 = 0.796

From ACI Cl. 10.3.2, ρ_{bal} = 0.03317

$$0.85 \times \beta_1 \times f_c' \times \left(\frac{600}{600 + f_y}\right)$$

From ACI Cl. 10.3.3, ρ_{max} = $0.75 \times \rho_{bal}$ = 0.02488

From ACI Cl.7.12.2, ρ_{min} = 0.00169

From Ref. 1, Eq. 3.8.4a, constant m = $\frac{F_y}{(0.85 \times F_c')}$ = 14.118

Calculate reinforcement ratio ρ for critical load case

Design for flexure about X axis is performed at the face of the column at a distance, D_z = $0.5 \times L \pm 0.5 \times B_{col} + O_{zd}$ = 1.025 m

Ultimate moment, $M_u|_{x=D_x}$ = 87.752 kNm

Nominal moment capacity, M_n = $\frac{M_u}{\Phi}$ = 97.502 kNm

(Based on effective depth) Required ρ = $\frac{1}{m} \times \left[1 - \sqrt{1 - 2 \times m \times \frac{M_n}{(F_y \times W \times d_{eff}^2)}}\right]$ = 0.00118

(Based on gross depth) ρ x d_{eff} / Depth = 0.00098

Since ρ= ρ_{min} ρ_{min} Governs

Area of Steel Required, A_s = $\rho \times W \times d_{eff}$ = 1217.111 mm2

Selected Bar Size = #12

Minimum spacing allowed (S_{min}) = 125.000mm

Selected spacing (S) = 150.000mm

S_{min}<= S <= S_{max} and selected bar size < selected maximum bar size...

The reinforcement is accepted.

Based on spacing reinforcement increment; provided reinforcement is

#12 @ 150mm o.c.

Required development length for bars = $\frac{d_b \times f_y}{20 \times \lambda \times \sqrt{f_c}}$ = =0.410 m

Available development length for bars, D_L = $0.5 \times (L - D_{col}) - C_{cover}$ = 0.725 m

Try bar size # 12 Area of one bar = 113.097 mm2

Number of bars required, N_{bar} = $\frac{A_s}{A_{bar}}$ = 13

Because the number of bars is rounded up, make sure new reinforcement ratio < ρ_{max}

Total reinforcement area, A_{s_total} = N_{bar} X (Area of one bar) = 1470.255 mm2

d_{eff} = D - C_{cover} - 1.5 X (dia. of one bar) = 0.332 m

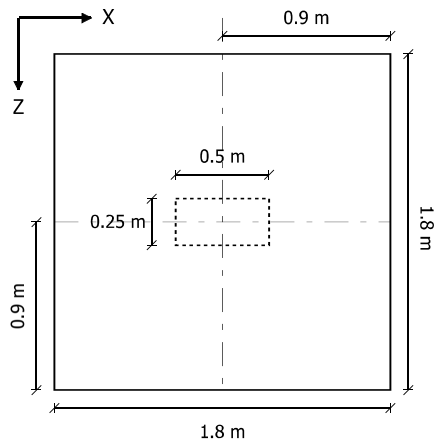
Reinforcement ratio, ρ = $\frac{A_{s_total}}{(d_{eff} \times W)}$ = 0.00246

From ACI Cl.7.6.1, minimum req'd clear distance between bars

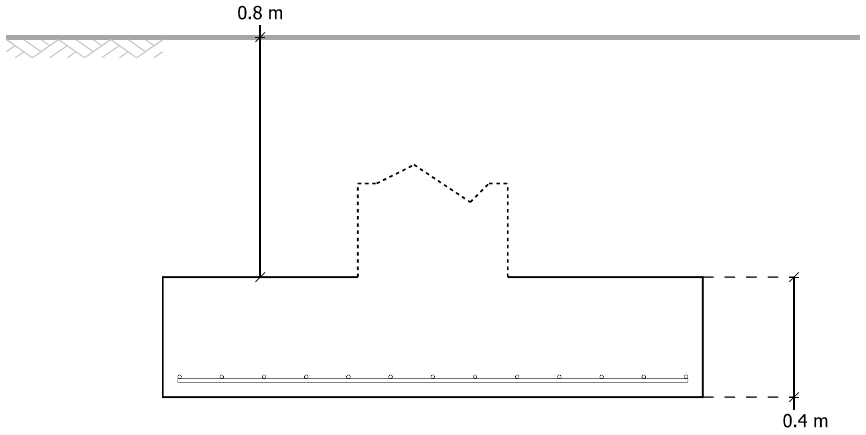
C_d = max (Diameter of one bar, 1.0" (25.4mm), Min. User Spacing) = 125.000mm

Provided Steel Area / Required Steel Area = 1.208

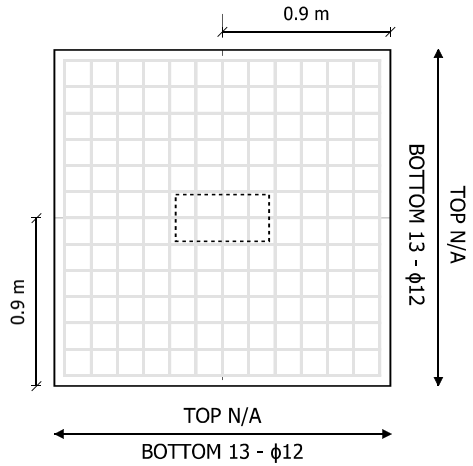
Check to see if width is sufficient to accomodate bars



Plan



Elevation



Plan

	Dimensions (m)			Reinforcement			
	L	B	D	AstX(T)	AstZ(T)	AstX(B)	AstZ(B)
Foundation	1.8 m	1.8 m	0.4 m	N/A	N/A	13 - ϕ12	13 - ϕ12
	l	b	h	Main Reinforcement		Trans Reinforcement	
Pedestal	N/A	N/A	N/A	N/A		N/A	

Print Calculation Sheet